

Kinematic Restoration Handbook

Kinematic Alignment
Inverse Kinematic Alignment

Kinematic Alignment with VELYS™ Robotic-Assisted Solution

T004
T004
Femur First Technique of resurfacing the femur based off registration then cutting the tibia with respect to surgeon priority:

- Resurfacing [Bone Cuts – mm]
- Long Leg Alignment [Target/Bounded HKA]
- Gap balanced

Anatomic Variability



■ KNEE

Coronal Plane Alignment of the Knee (CPAK) classification

A NEW SYSTEM FOR DESCRIBING KNEE PHENOTYPES

S. J. MacDessi,
W. Griffiths-Jones,
I. A. Harris,
J. Bellemans,
D. B. Chen

From Sydney Knee
Specialists, Sydney,
Australia

Aims

A comprehensive classification for coronal lower limb alignment with predictive capabilities for knee balance would be beneficial in total knee arthroplasty (TKA). This paper describes the Coronal Plane Alignment of the Knee (CPAK) classification and examines its utility in preoperative soft tissue balance prediction, comparing kinematic alignment (KA) to mechanical alignment (MA).

Methods

A radiological analysis of 500 healthy and 500 osteoarthritic (OA) knees was used to assess the applicability of the CPAK classification. CPAK comprises nine phenotypes based on the arithmetic HKA (aHKA) that estimates constitutional limb alignment and joint line obliqui-

Reference: MacDessi SJ, Griffiths-Jones W, Harris IA, Bellemans J, Chen DB. Coronal Plane Alignment of the Knee (CPAK) classification. Bone Joint J. 2021;103-B(2):329-337. doi:10.1302/0301-620X.103B2.BJJ-2020-1050.R1

Joint Line Obliquity (MPTA + LDFA)

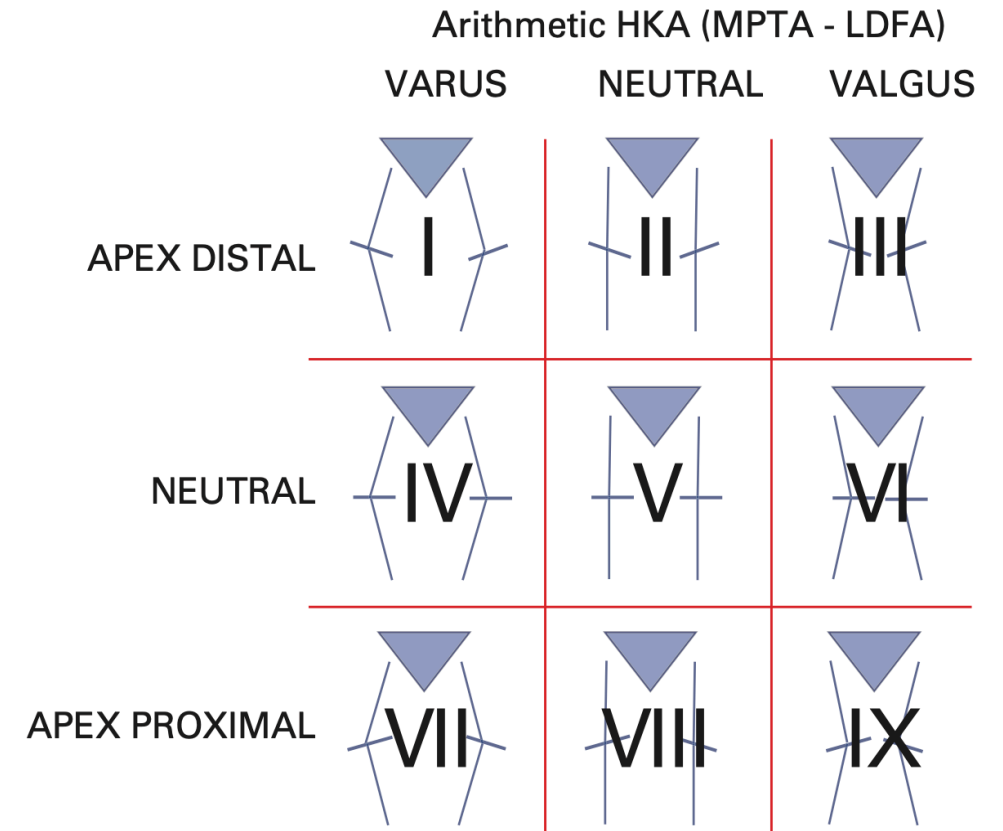
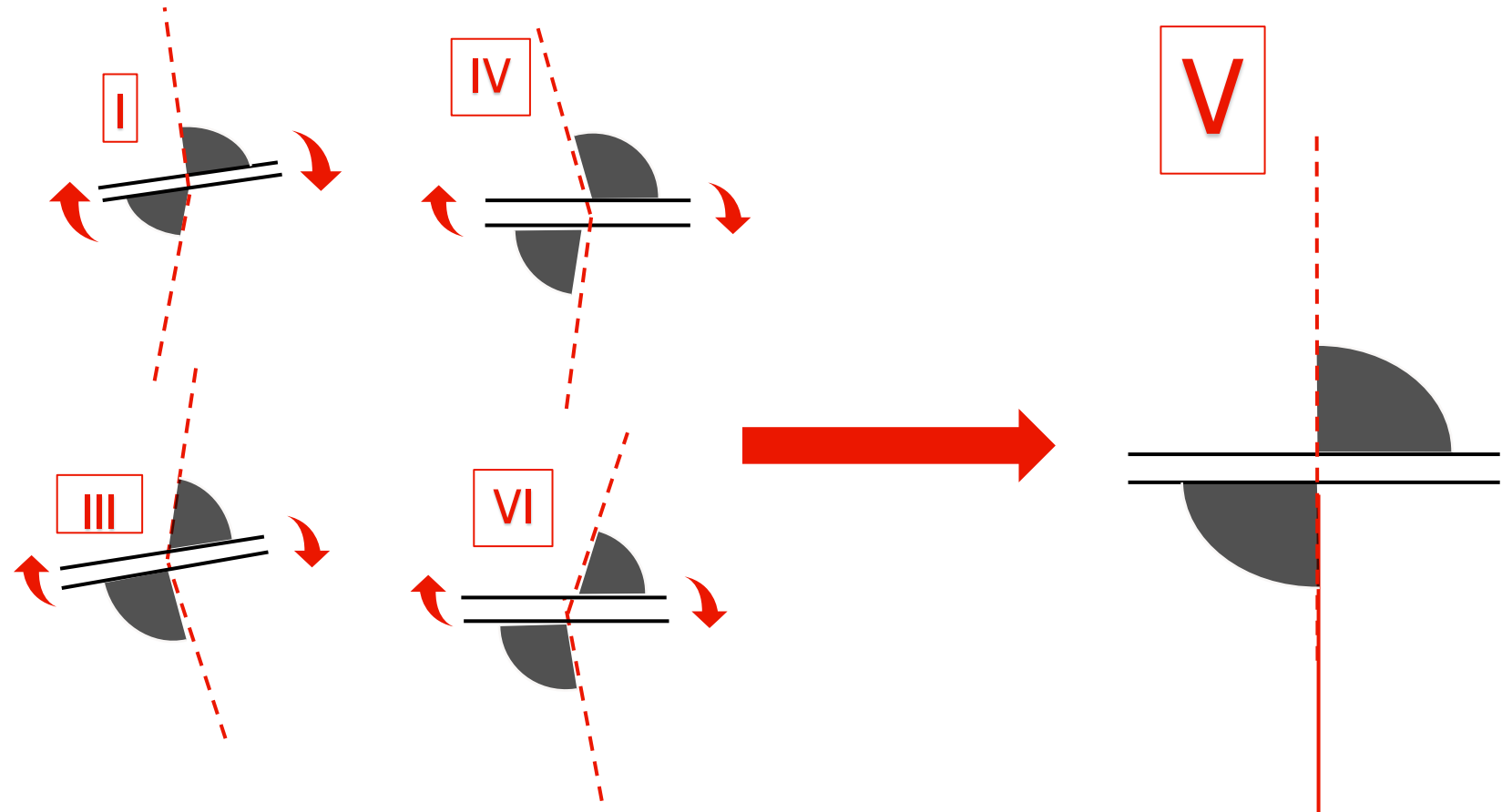
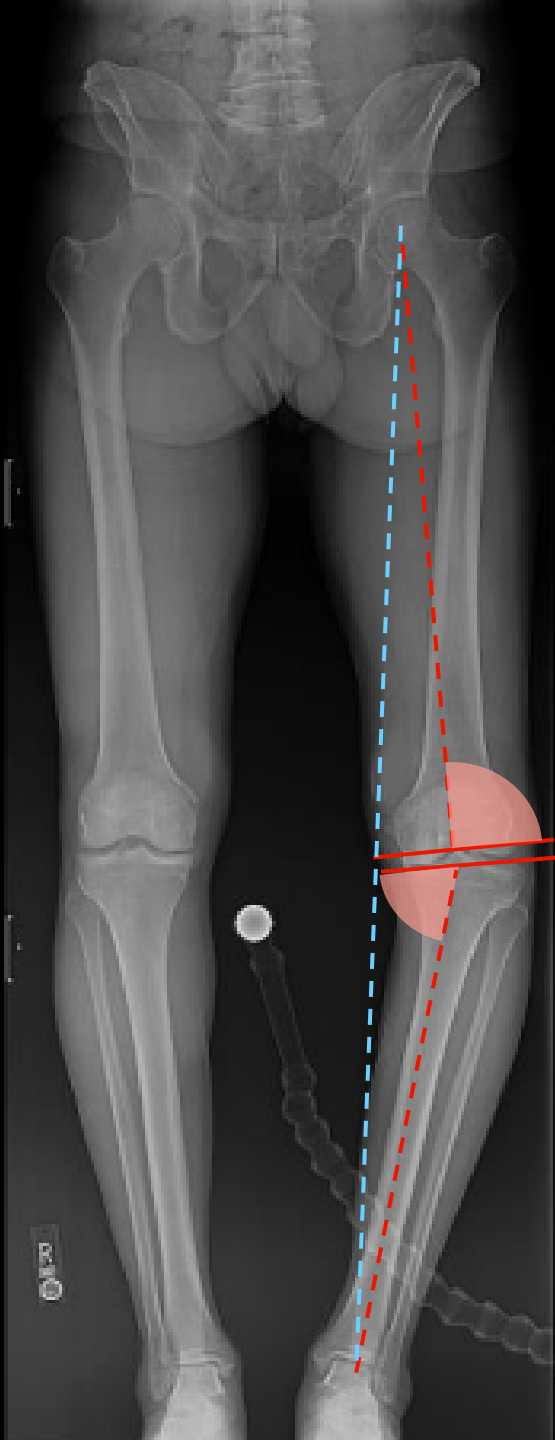


Fig. 4

Coronal Plane Alignment of the Knee classification (CPAK) with nine theoretical types of knee. Arithmetic HKA, arithmetic hip-knee-ankle angle.

Historical techniques change native knee phenotypes



Native CPAK

Final CPAK

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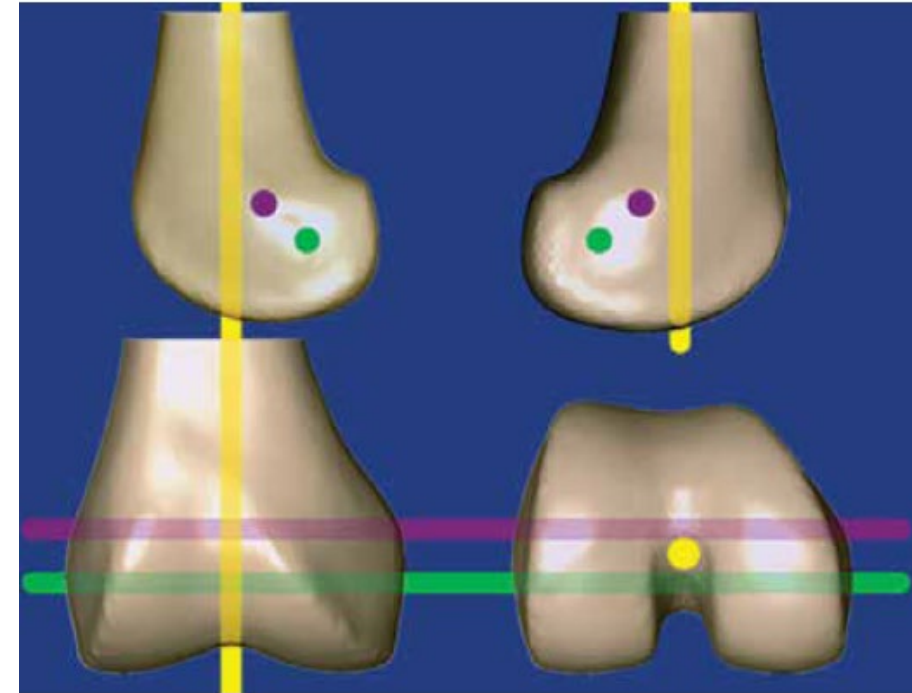
Kinematic Alignment Defined

Goal: Restore the natural difference in symmetry and varus-valgus laxity between 0° extension and 90° of flexion in the normal knee (Howell KA defined)

- **Restore normal kinematics in TKA.**
 - Interaction between ligaments, menisci, and articular surfaces of femur, tibia, and patella determine normal knee kinematics...and are described by 3 axes.

How: Surgeon strives to position femoral and tibial components to resurface the knee and restore the 3 natural joint lines (axes) in the arthritic knee

- Green line – indicates **transverse axis** in the femur about which the **tibia flexes and extends**
- Magenta line – indicates the **transverse axis** in the femur about which **patella flexes and extends**
- Yellow line – indicates the **longitudinal axis** in tibia **about which the tibia internally-externally rotates** on femur



Kinematic Alignment Defined – how to recreate 3 axes¹

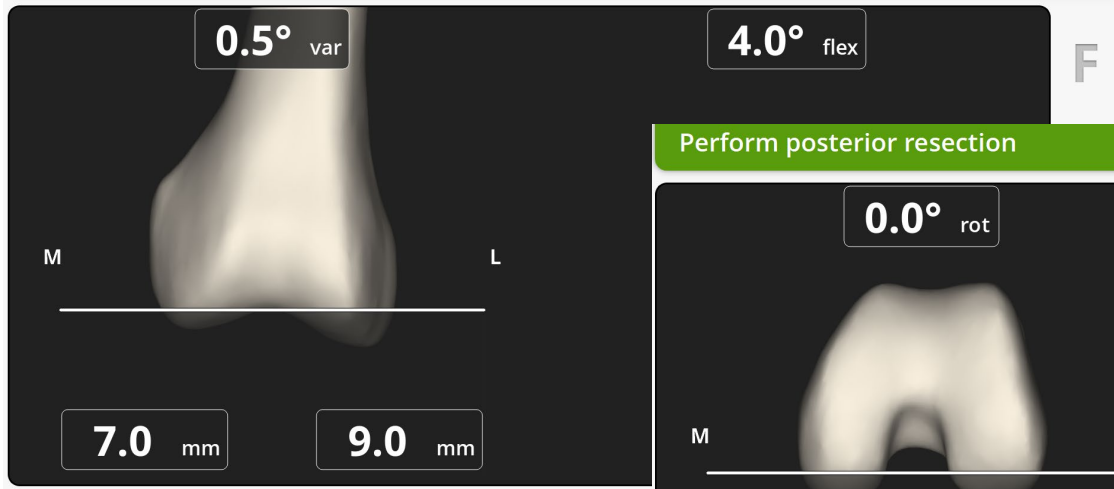
Resurface Femur and Tibia to Pre-diseased state

Step 1: Resurface Distal Femur in coronal plane to recreate lateral distal femoral angle

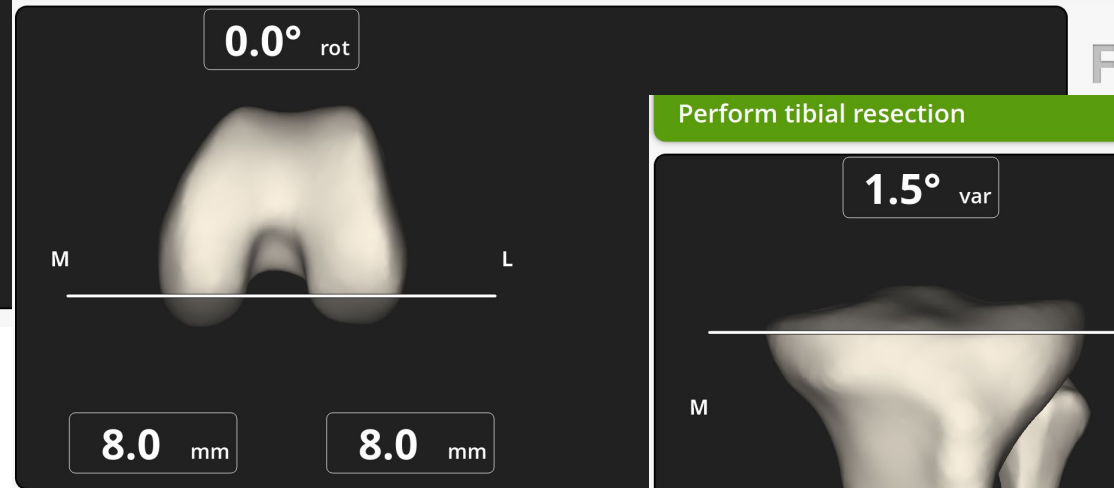
Step 2: Resurface Posterior Condyles to restore native posterior condylar offset

Step 3: Resurface Proximal Tibia to recreate medial proximal tibial angle and slope

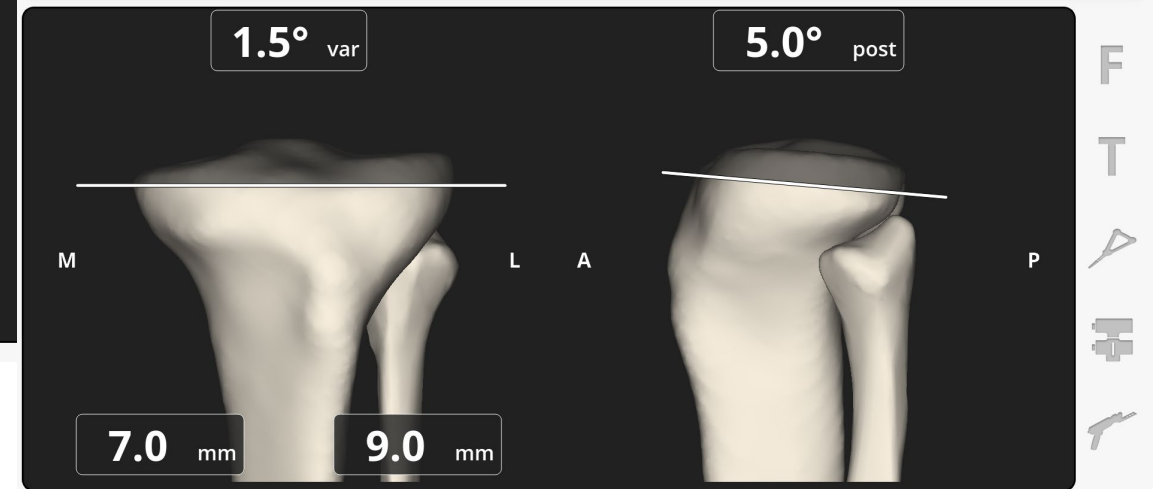
Perform distal resection



Perform posterior resection



Perform tibial resection



NOT FOR CLINICAL USE

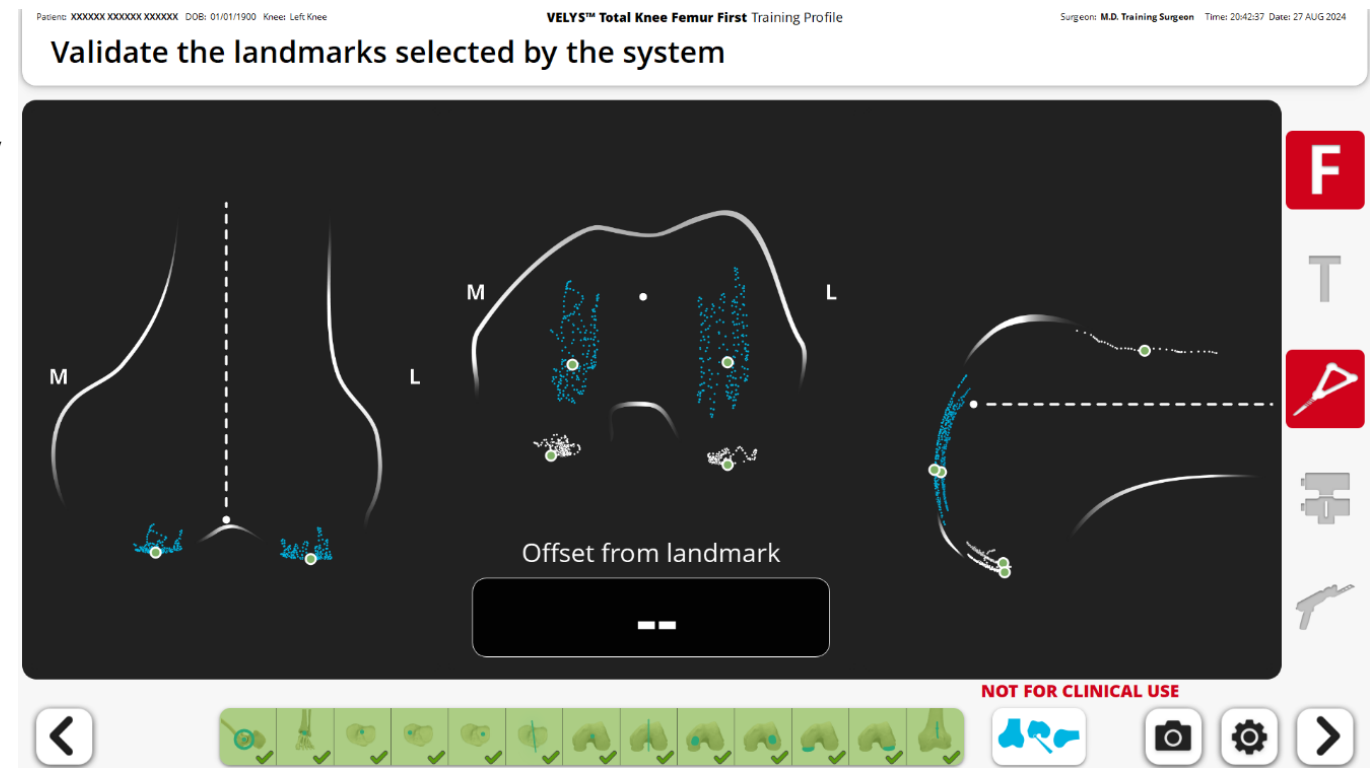
Landmark Acquisition – surgeon defined¹

Surgeons dictate coronal plane resection depths and alignment through user defined registration

- Image based systems don't give the surgeon freedom to define origin of resection depths
 - CT scans must be segmented by third party

Surgeons will have ability to accommodate soft and hard tissue at their choosing

Surgeons can validate femoral registration up to 0.5 mm of accuracy



Landmark Acquisition Technique¹

Define Femoral coordinate system

Femoral Head Center

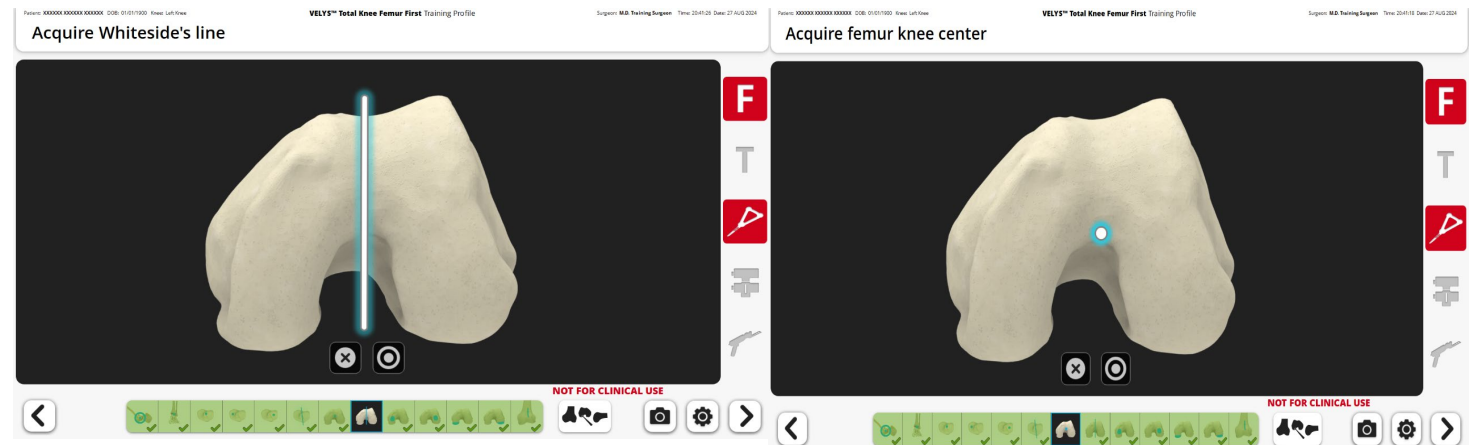
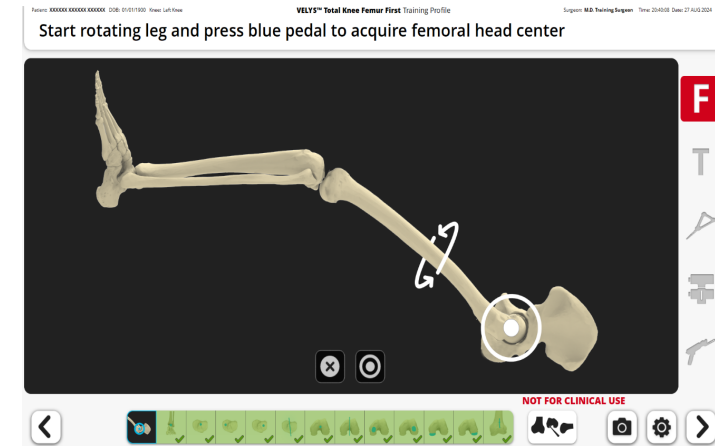
- Define proximal extent of femoral mechanical axis

Femoral Knee Center

- Define distal extent of femoral mechanical axis
- Apex of intercondylar notch

Whitesides Line (PCA 3 ER)

- Define secondary Int/Ext rotation reference for sagittal alignment



Landmark Acquisition Technique¹

Define Tibial coordinate system

Medial & Lateral Malleoli

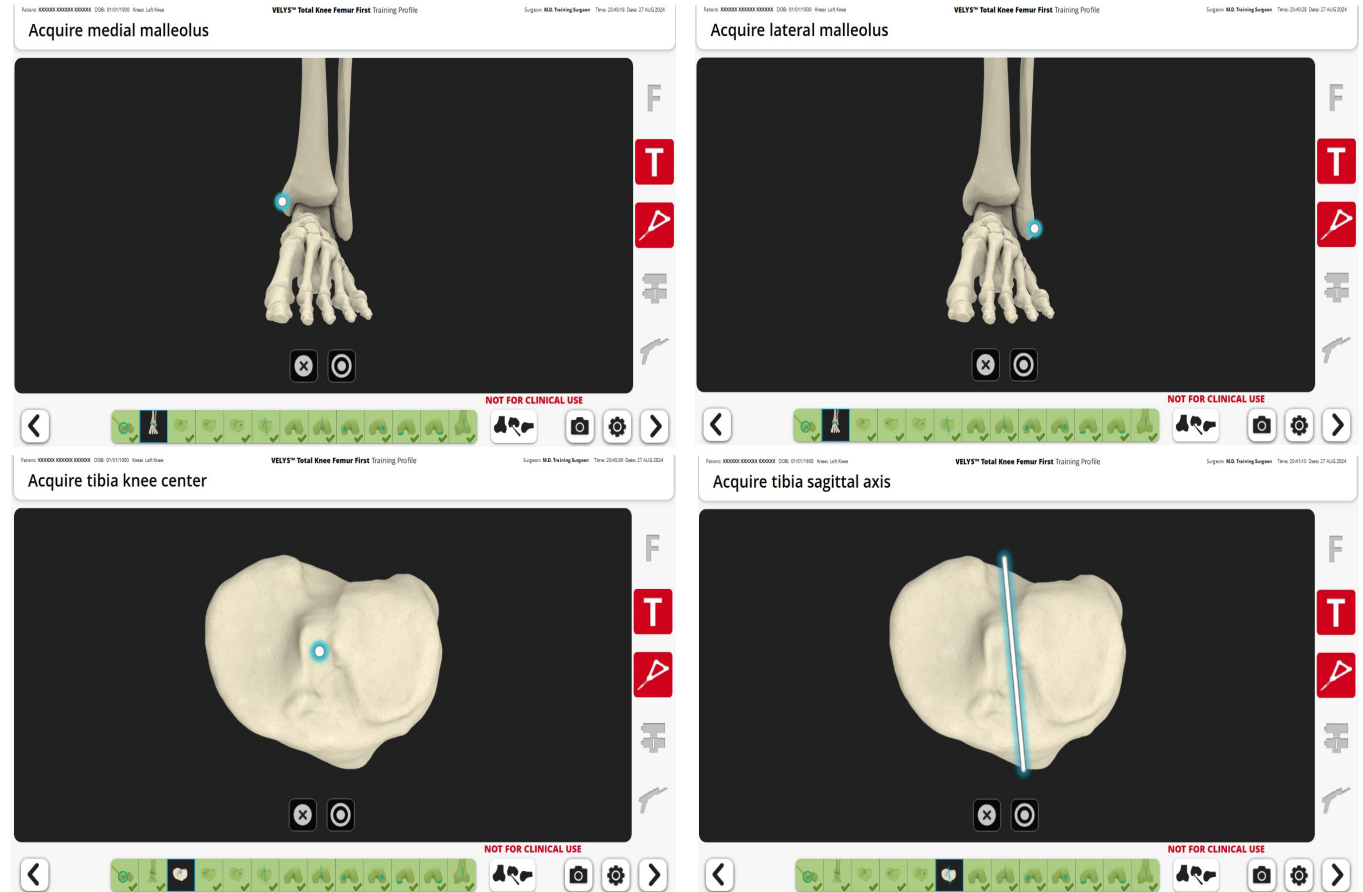
- Define distal extent of Tibial mechanical axis
- Palpate most prominent points on each malleoli

Tibial Knee Center

- Define proximal extent of Tibial mechanical axis
- Mid sagittal / Mid coronal in ACL footprint

Tibial Sagittal Axis

- Tip of the pointer aligned to PCL insertion posteriorly
- Shaft of pointer bisects tibial spine
- Aligns parallel to A/P axis of lateral plateau so that Sagittal Axis is perpendicular to F/E axis of femur and parallel to F/E plane

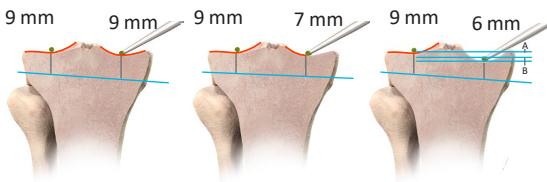
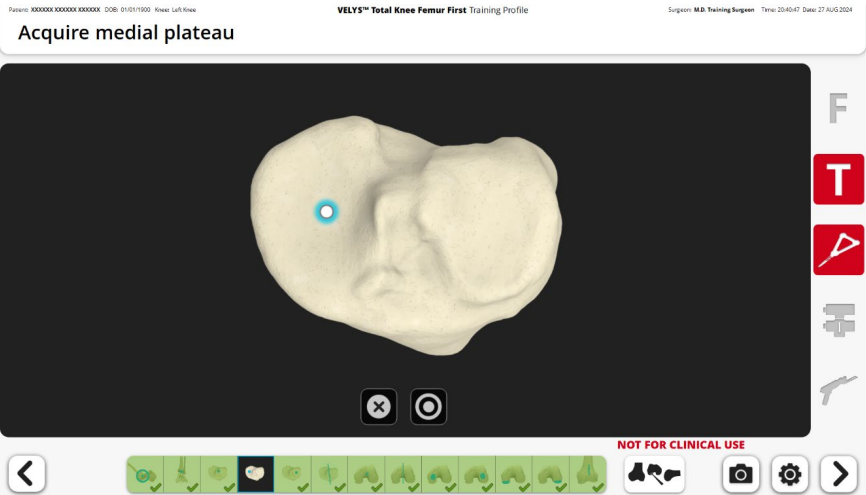


Landmark Acquisition Technique¹

Define Tibial Resection Reference Points [varus knee]

Medial Plateau

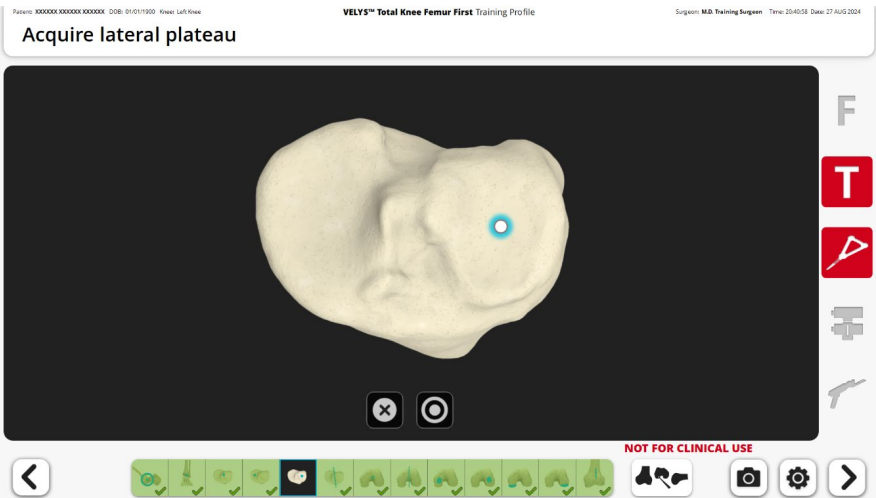
- Assess cartilage and bone loss/wear pattern
- Acquire medial resection reference at base of tibial spine
- Surgeon has flexibility to define where best to acquire this point based on their own opinion



Scenario	1	2	3
Cartilage loss	0 mm	2 mm	2 mm
Bone loss	0 mm	0 mm	1 mm
Resection	9 mm	7 mm	6 mm
Total	9 mm	9 mm	9 mm

Lateral Plateau

- Assess unworn plateau
- Acquire lateral resection reference at base of tibial spine
- Surgeon has flexibility to define where best to acquire this point based on their own opinion

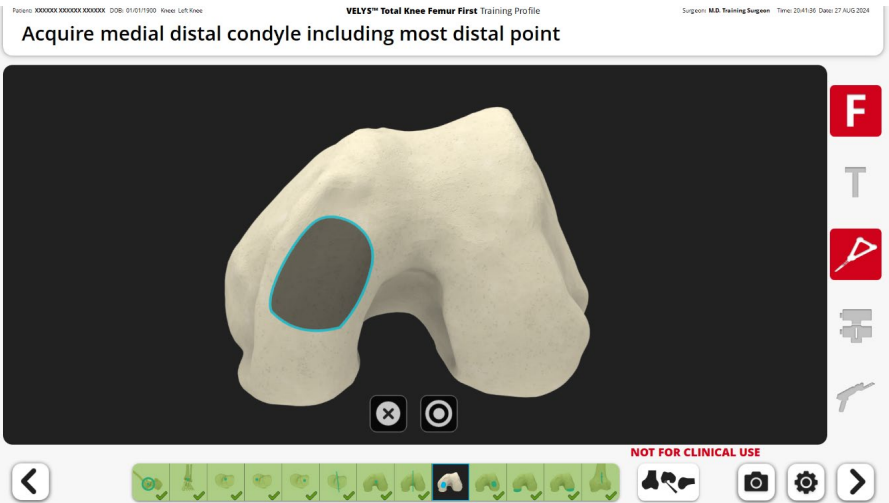


Landmark Acquisition Technique¹

Define Coronal Plane [varus knee]

Medial Distal Femoral Condyle

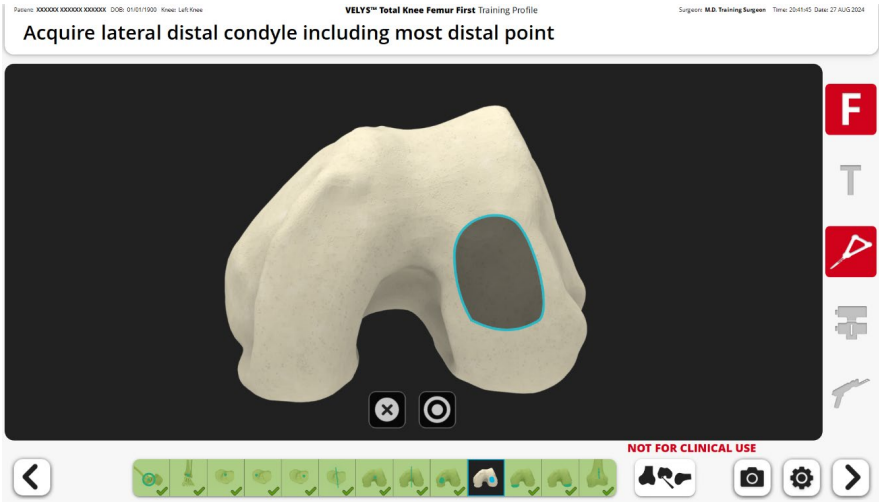
- Define most distal point on medial femoral condyle
- Assess cartilage loss/wear pattern
- Acquire most distal medial femoral condyle point through painting broad surface area



Scenario	1	2	3
Cartilage loss	0 mm	2 mm	2 mm
Bone loss	0 mm	0 mm	1 mm
Resection	9 mm	7 mm	6 mm
Total	9 mm	9 mm	9 mm

Lateral Distal Femoral Condyle

- Define most distal point on lateral femoral condyle
- Acquire most distal lateral femoral condyle point through painting broad surface area

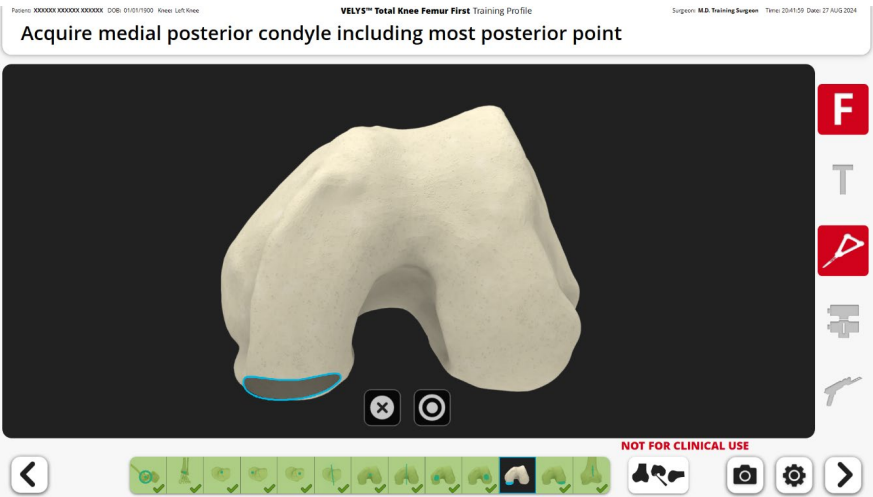


Landmark Acquisition Technique¹

Define Femoral Sagittal Plane [varus knee]

Medial Posterior Femoral Condyle

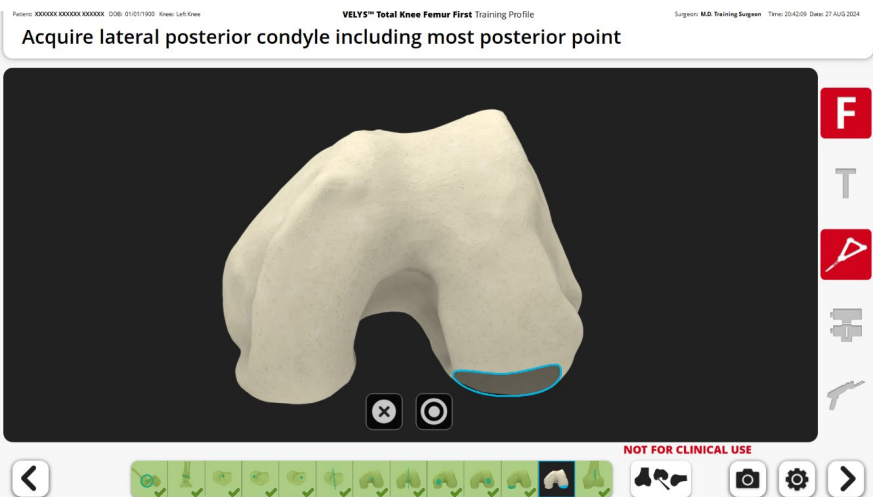
- Define most posterior medial femoral condyle point
- Assess cartilage loss/wear pattern
- Acquire most posterior medial point through surface acquisition



	8 mm	8 mm	8 mm	6 mm	8 mm	5 mm
Scenario	1	2	3			
Cartilage loss	0 mm	2 mm	2 mm			
Bone loss	0 mm	0 mm	1 mm			
Resection	8 mm	6 mm	5 mm			
Total	8 mm	8 mm	8 mm			

Lateral Posterior Femoral Condyle

- Define most posterior lateral femoral condyle point
- Assess unworn posterior condyle
- Acquire most posterior lateral point through surface acquisition

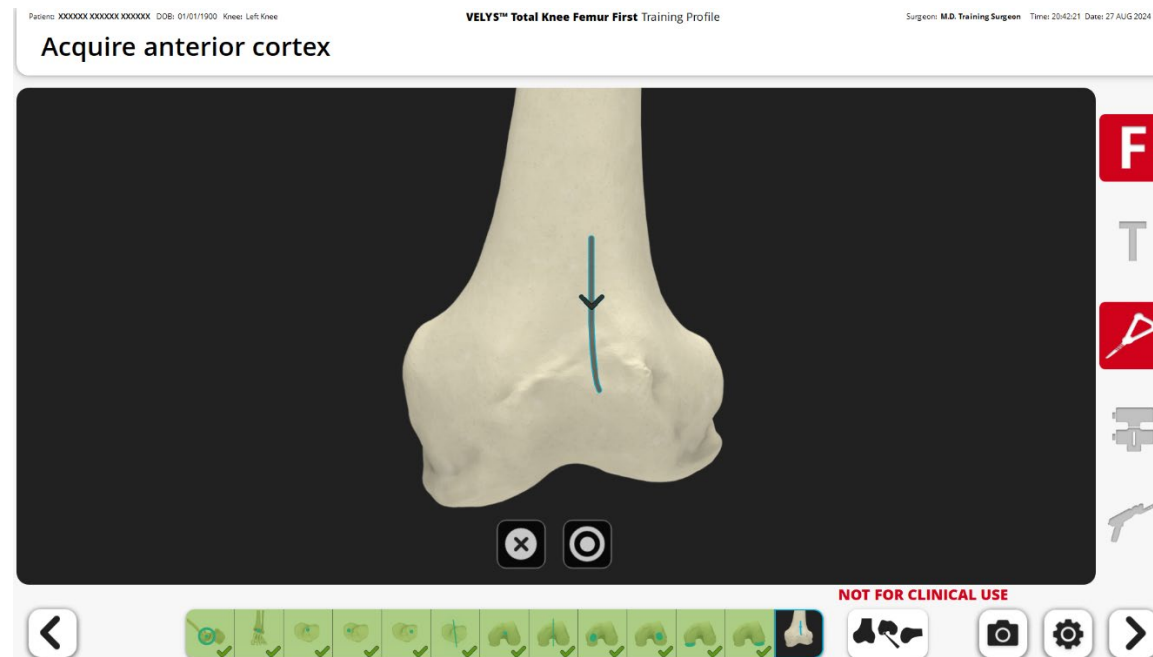


Landmark Acquisition Technique¹

Define Femoral Sizing and Saw Blade Exit Point [varus knee]

Anterior Cortex Line

- Define femoral sizing and saw blade exit point
- Acquire anterior cortex line by running pointer in a straight line down the anterior cortex of the femur
 - Start pointer more proximal and run pointer down cortex about halfway up the lateral ridge

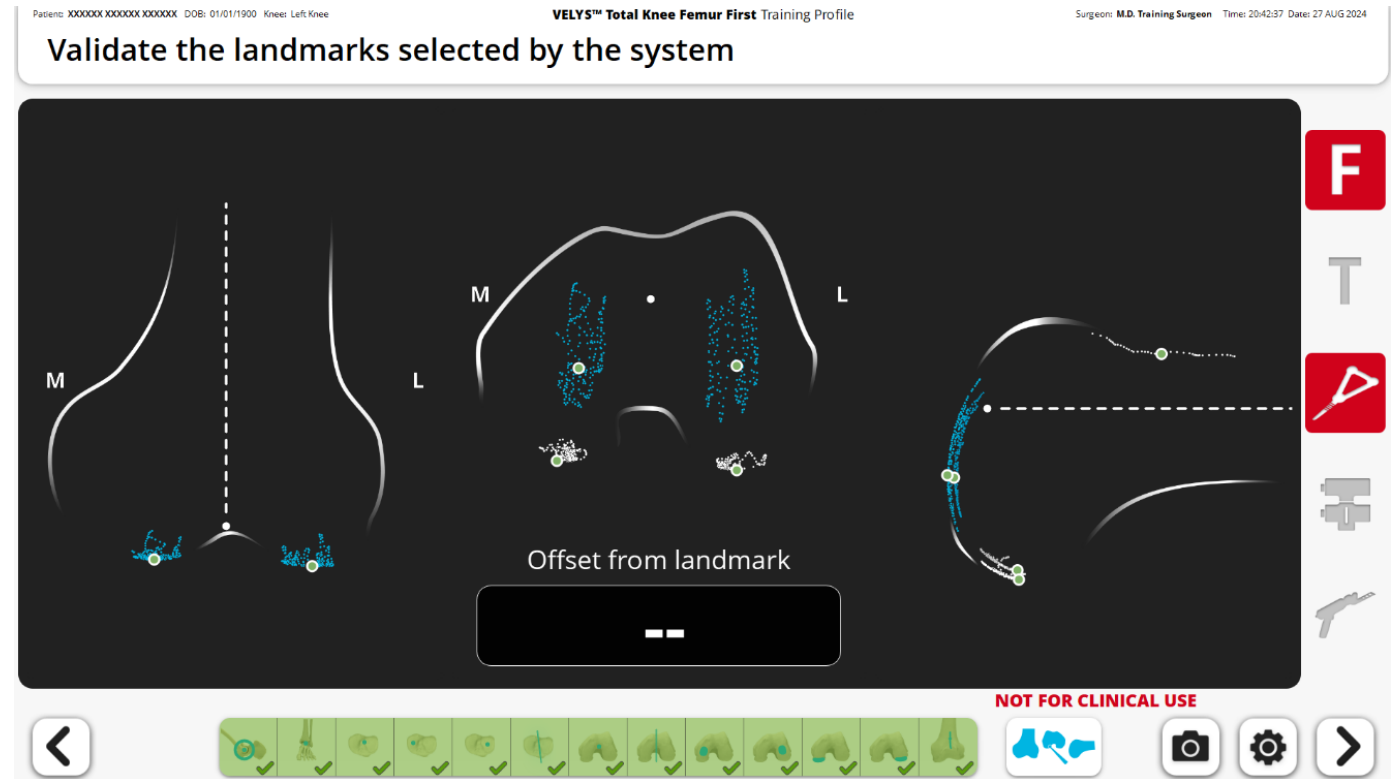


Landmark Acquisition Technique¹

Validate Femoral Landmarks [varus knee]

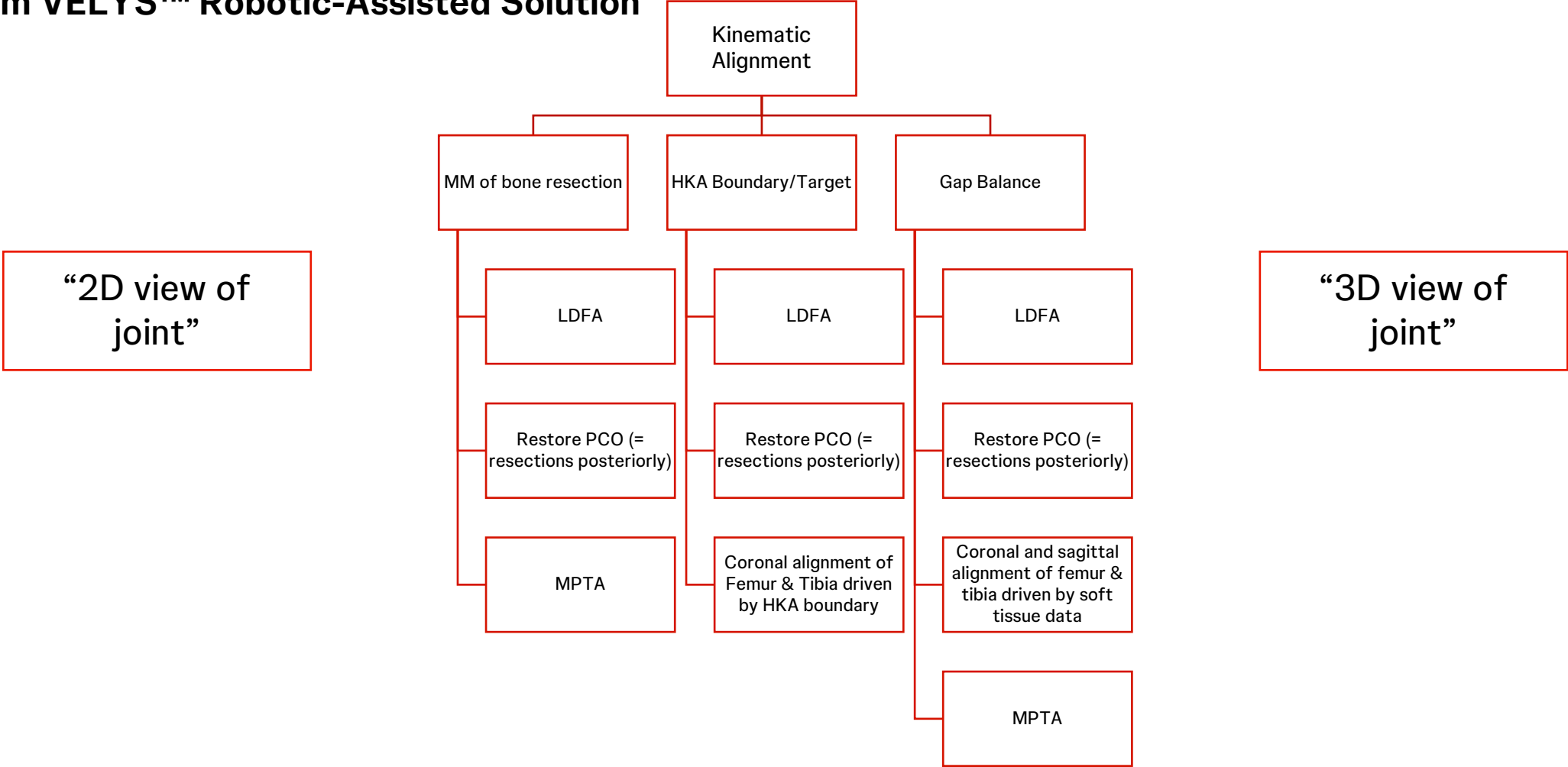
Surgeon defined landmark acquisition validation

- Confirm landmarks used to define femoral alignment in coronal plane (ext/flex)
- Surgeon has ability to verify accuracy up to 0.5mm
- If surgeon can't verify, surgeon has ability to re-acquire landmark
- Surgeon does NOT have to compromise their landmark acquisition based on pre-op imaging



Kinematic Alignment Prioritization¹

Prioritizing mm of bone resection is acceptable, but there could be more information gleaned from VELYS™ Robotic-Assisted Solution

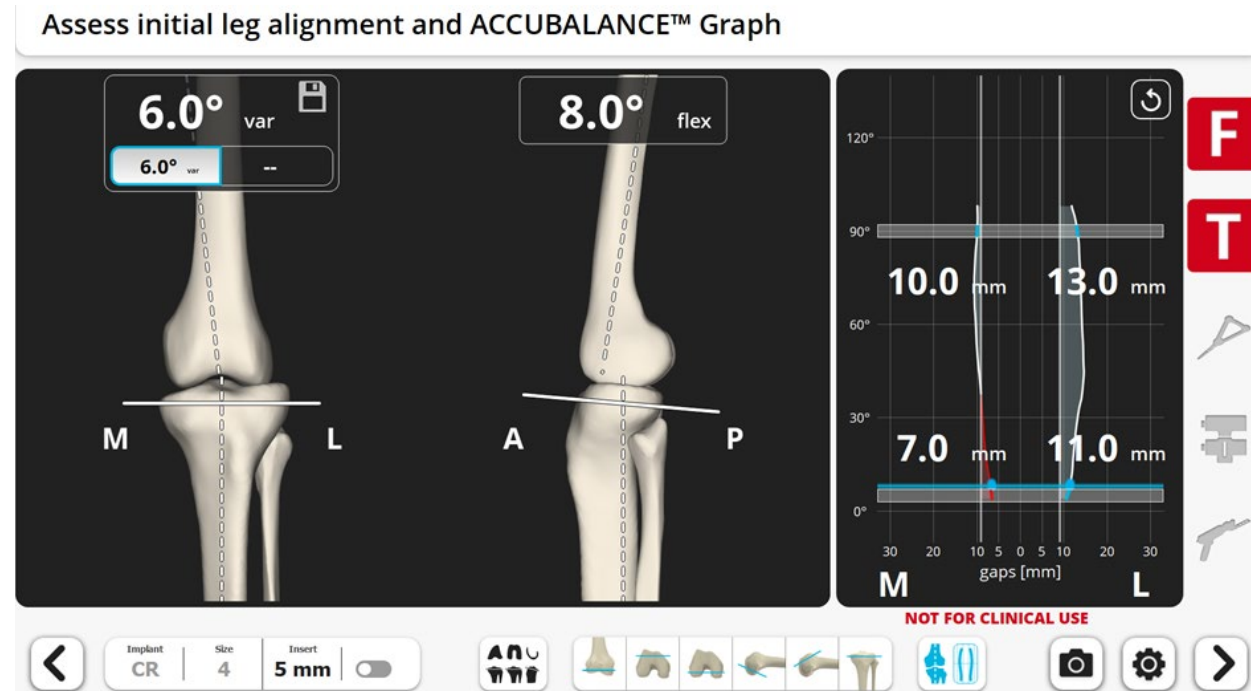


Initial Leg Alignment & ACCUBALANCE™ Graph¹

Priority – Resurfacing [bone cuts mm]

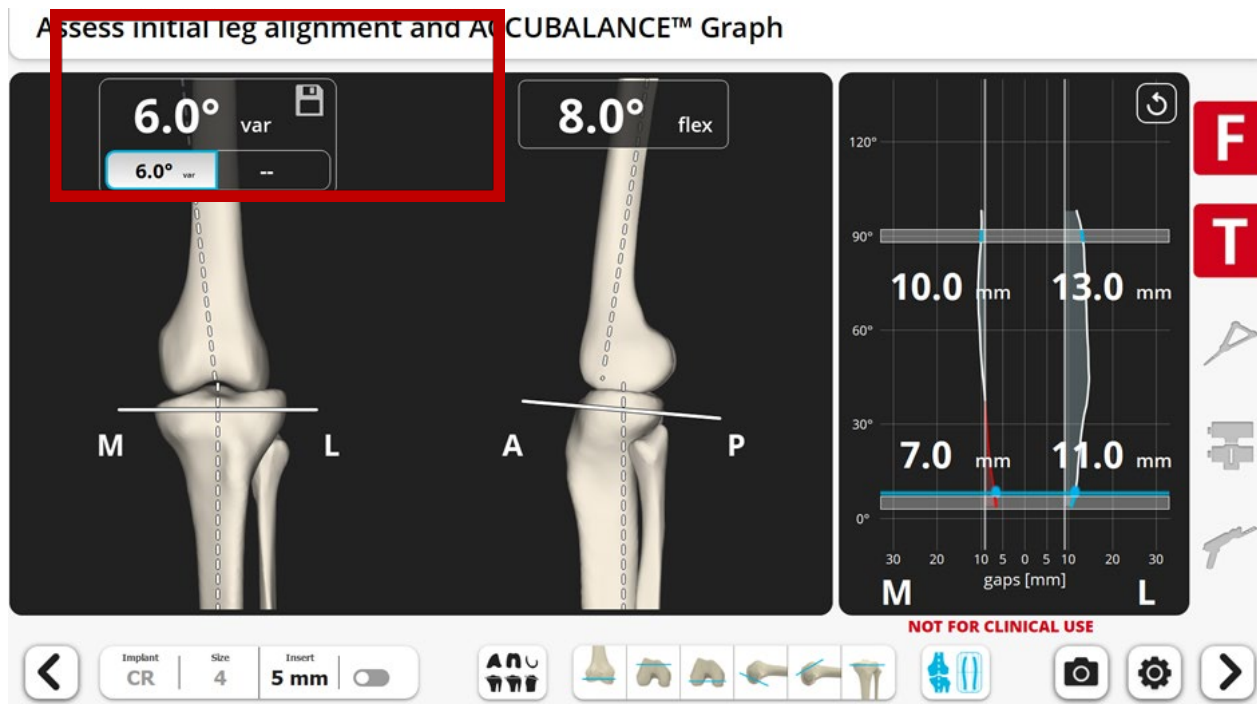
Femur First Technique of resurfacing the femur based off registration then cutting the tibia with respect to surgeon priority:

- Resurfacing [Bone Cuts – mm]
 - Surgeons who prioritize resurfacing will deprioritize this step
 - Surgeons can use this step to help confirm tibial sagittal plane is defined in appropriate direction
 - Femoral rotation in sagittal view
 - Location of femoral head w/ leg in flexion



Initial Leg Alignment & ACCUBALANCE™ Graph¹

Long Leg Alignment Target / Boundary



rHKA = 6° Varus

sHKA = 2° Varus

Definitions:

- **mHKA** – mechanical HKA
- **rHKA** – “resting” HKA where the leg is in full extension with no force applied (aka “deformity”)
- **sHKA** – “stressed” HKA where the leg corrects to when stressed to reverse direction as deformity (aka “**correctability**”)

Technique:

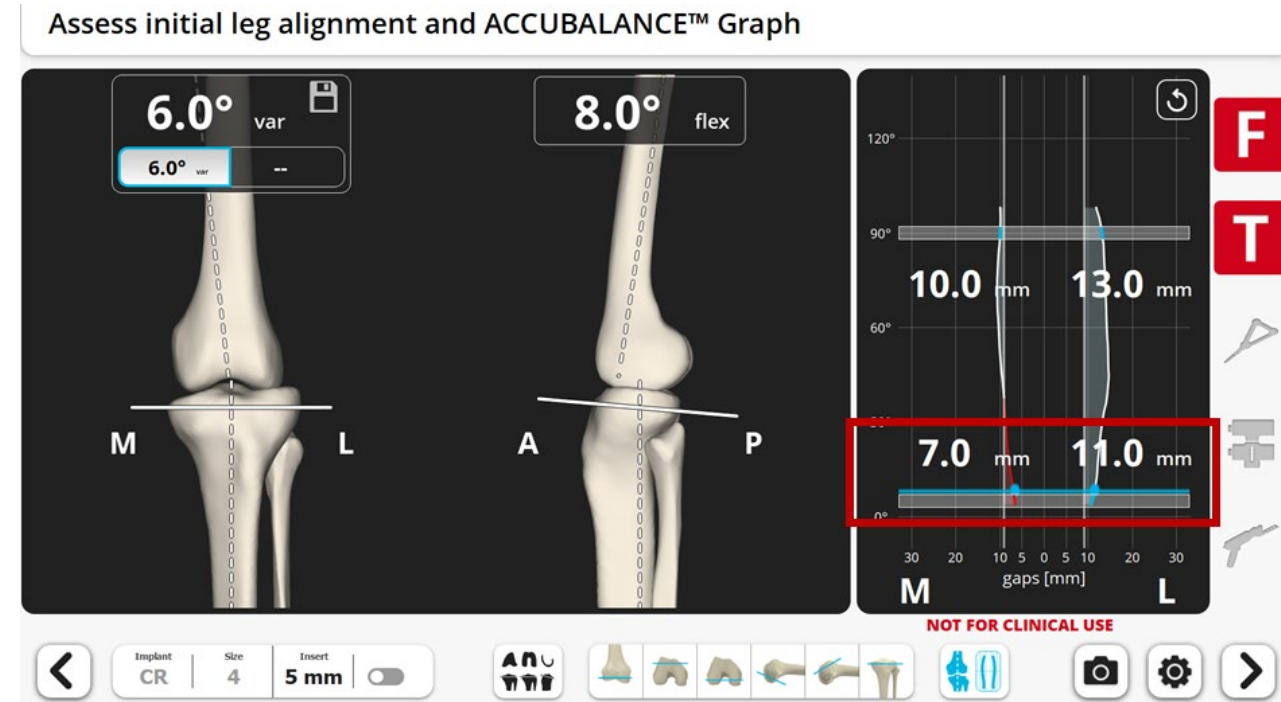
- With leg in full extension assess **rHKA**
- Apply corrected stress (**sHKA**) to leg in full extension
- sHKA becomes target HKA on ProAdjust Planning page and informs surgeon on coronal alignment of tibia

Initial Leg Alignment & ACCUBALANCE™ Graph¹

Priority – Gap Balanced

Femur First Technique of resurfacing the femur based off registration then cutting the tibia with respect to Extension Gap soft tissue balance (deformity):

- Gap balanced [Extension Gap]
 - Apply physiologic dynamic stress from patient's native terminal extension through ROM
 - Evaluate extension gap values and “digital signature” of ACCUBALANCE™ graph
 - Utilize Extension Gap Window to define extension gap data [identify where graph transitions from red to white line]
 - Posterior capsule and screw home mechanism disengaged
 - Extension gap balance helps surgeon inform coronal alignment on planning page



Priority – Resurfacing [bone cuts (mm)]¹

cut tibia to resurface based on registered landmarks

Resurface the femur:

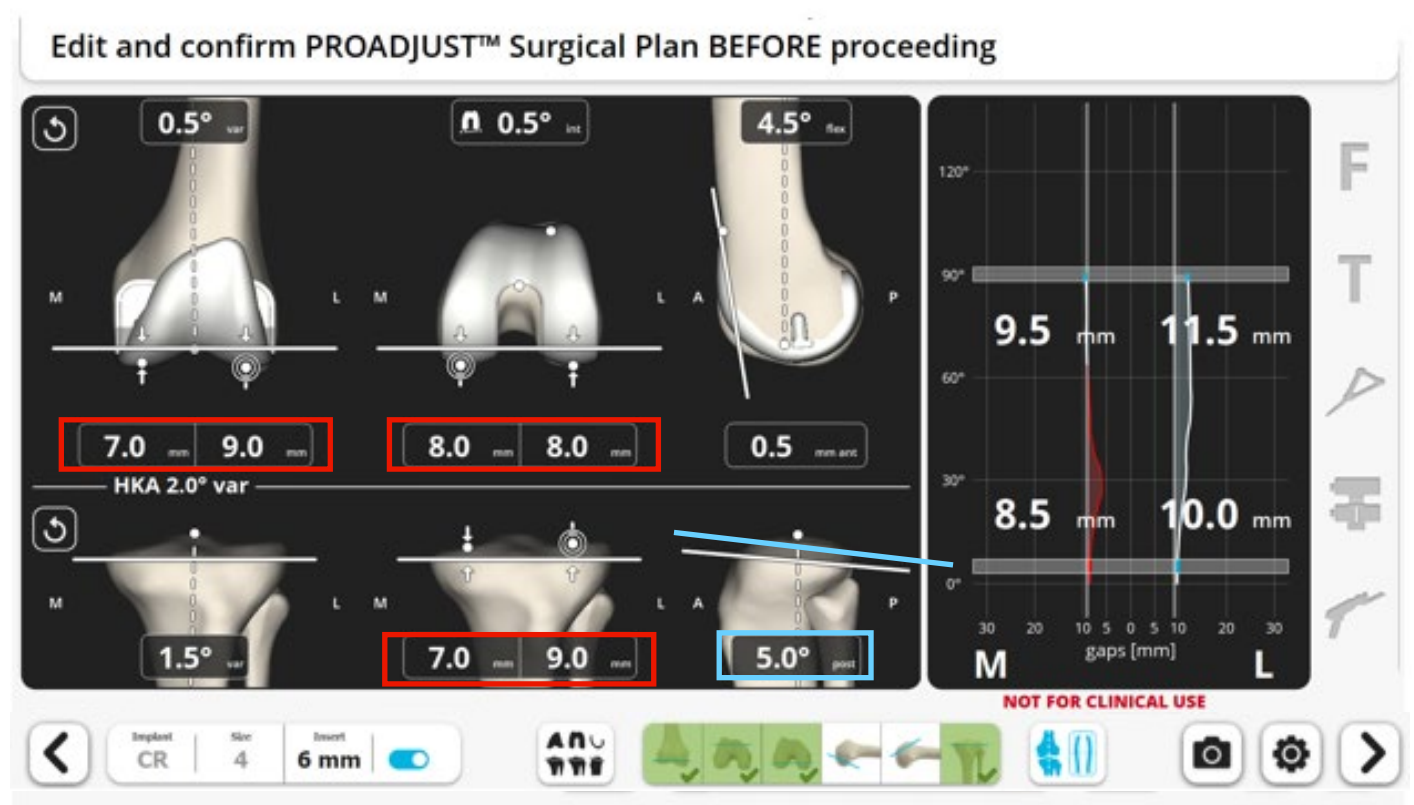
- resection depth determined by registration – bone or cartilage
- V/V function of bone cuts

Cut tibia to resurface the bone

- resection depth determined by registration – bone or cartilage
- V/V function of bone cuts
- Surgeon can define **patient specific slope** with **pointer**

V1.8 Resection Locks

- Utilize resection locks to avoid femoral & tibial resection depth compromise



Priority = Long Leg Alignment [HKA]¹

cut tibia to achieve a HKA Target / Boundary

Resurface the femur:

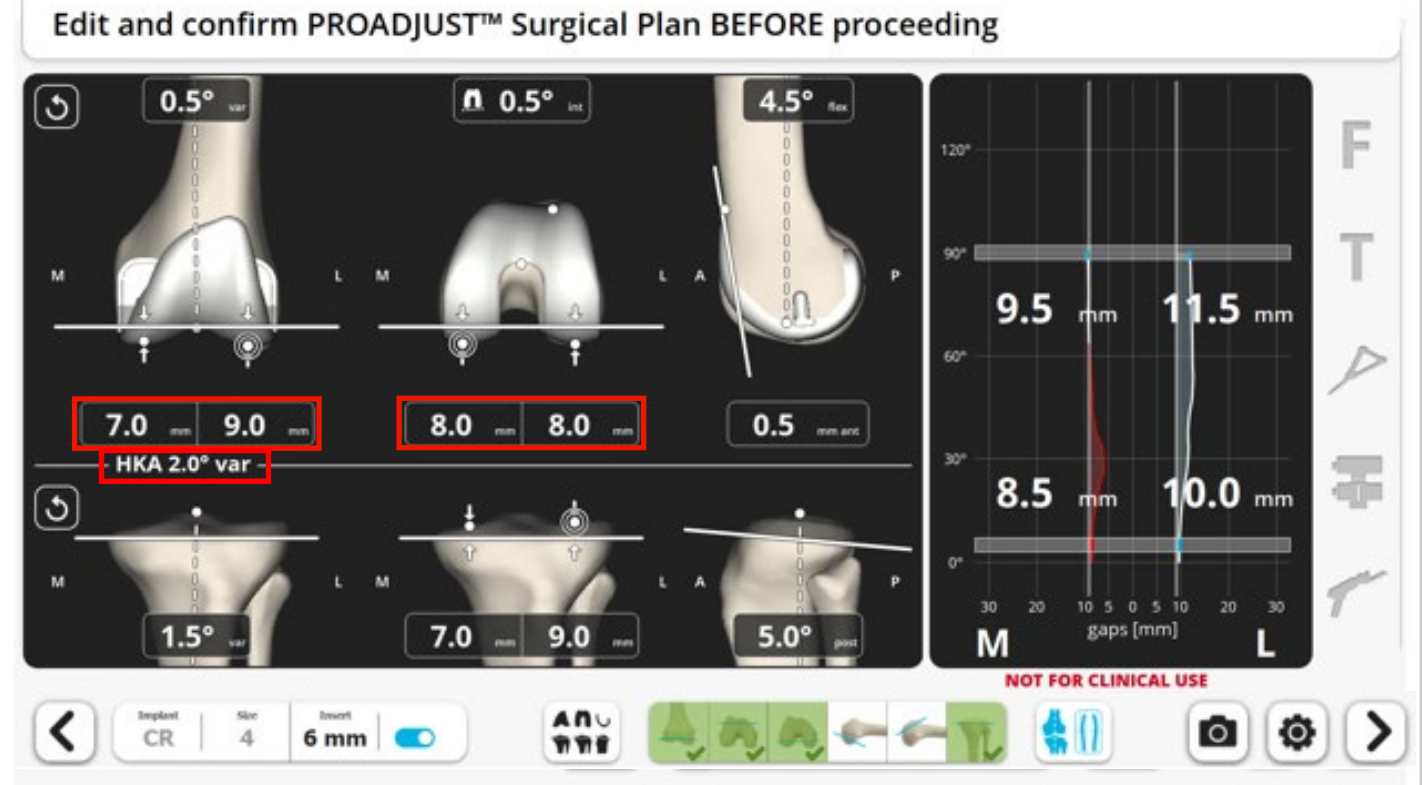
- resection depth determined by registration – bone or cartilage
- V/V function of bone cuts

Cut tibia to achieve target HKA (2° var)

- tibia V/V function of target HKA
- tibia resection depth function of V/V

V1.8 Resection Locks

- Utilize resection locks on screen to avoid coronal, sagittal, and long leg alignment compromise



rHKA = 6° Varus

sHKA = 2° Varus

Priority – Gap Balanced¹

cut tibia to balance extension space (and flexion gaps if surgeon uses boundaries) –
effectively translating distal femoral resection distally to create rectangular extension space

Resurface the femur:

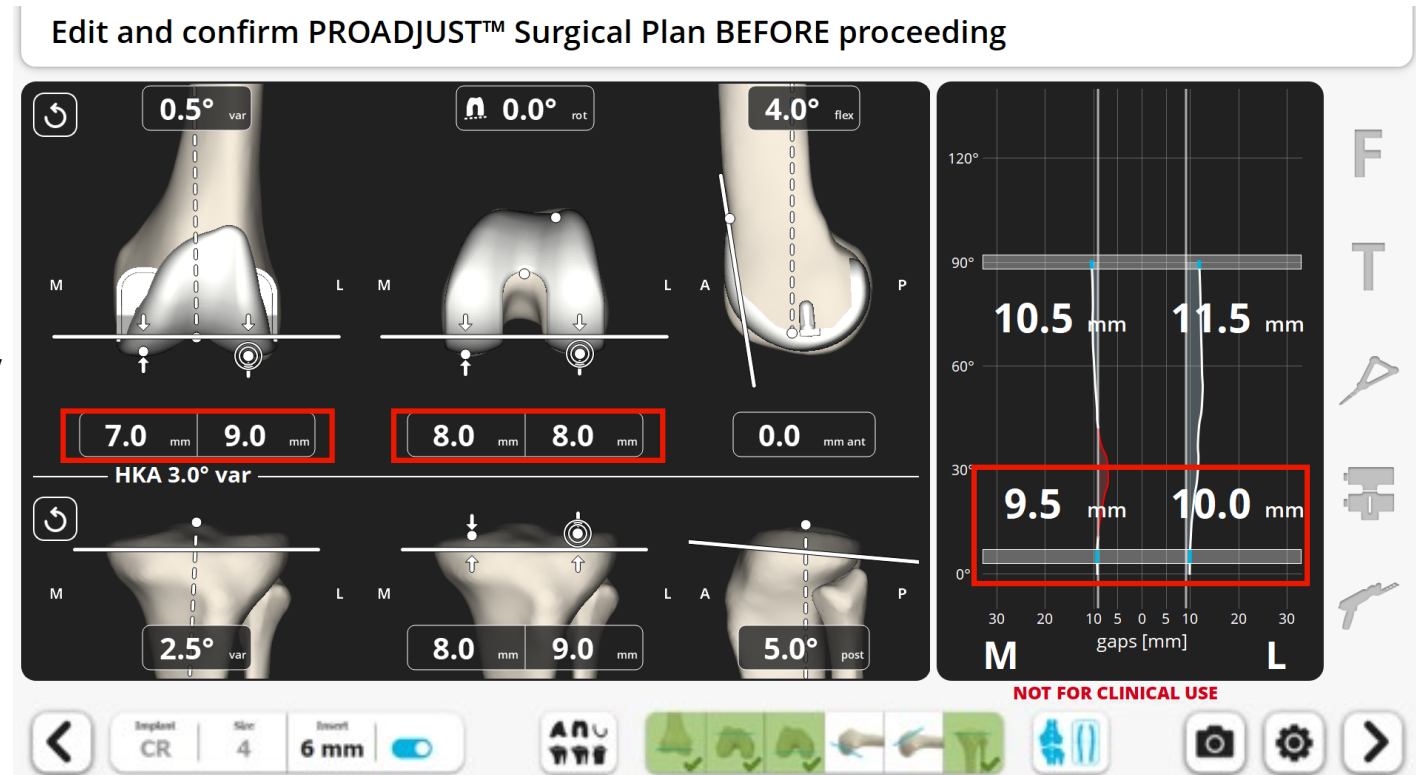
- resection depth determined by registration – bone or cartilage
- V/V function of bone cuts

Cut tibia to balance extension space

- With appropriately tensioned extension gap cut V / V into tibia to balance
- bone cuts & V/V function of balance

V1.8 Resection Locks

- Utilize resection locks on screen to avoid coronal & sagittal alignment and extension gap balance target compromise



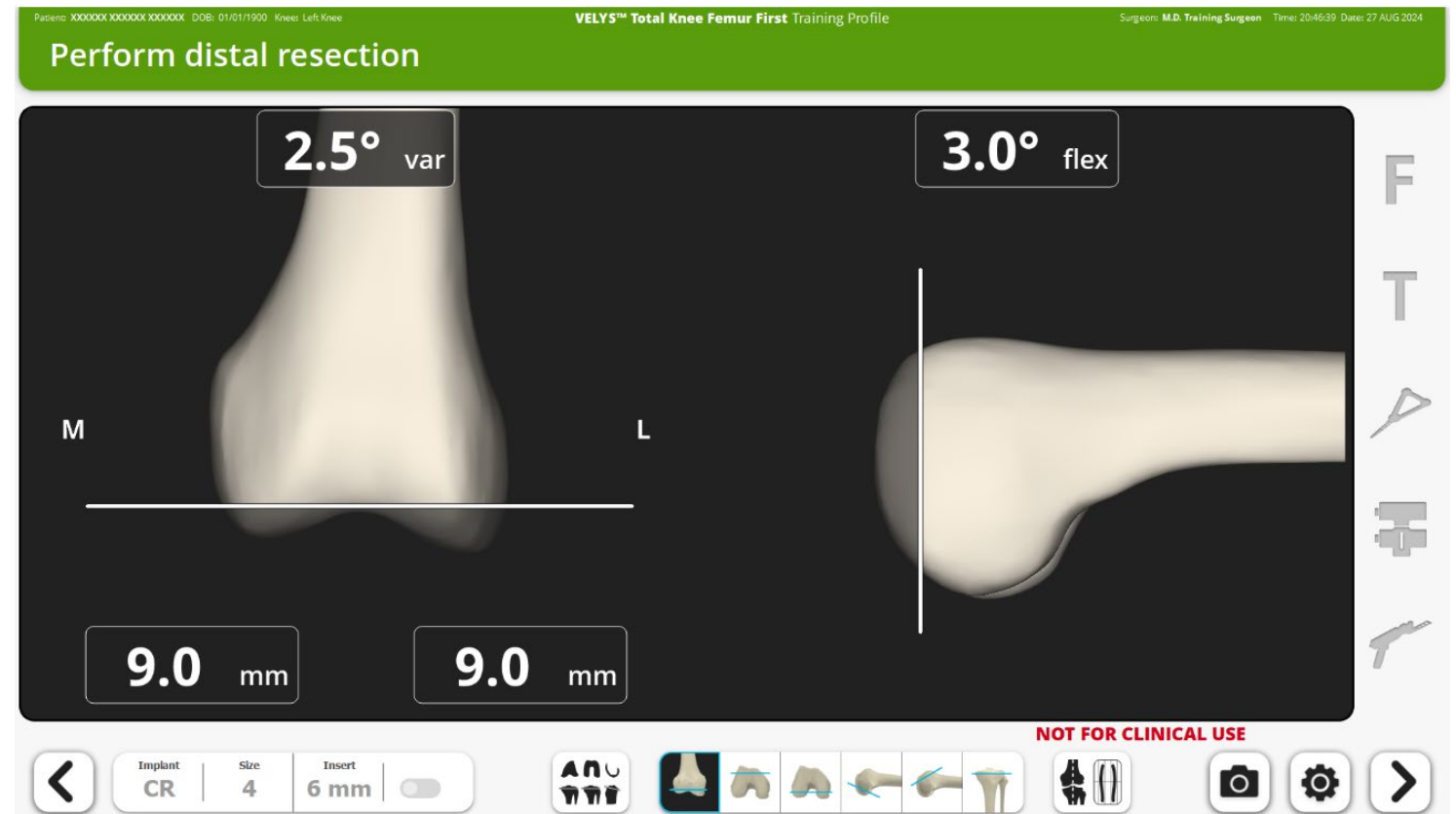
Plan Execution¹

Distal Cut [varus knee]

Execute distal femoral cut

Ensure accurate resection depth up to 0.5mm of accuracy by running the pointer along cut surface

Based on measurements and pointer verification, make any additional recuts to the distal femoral condyle cuts



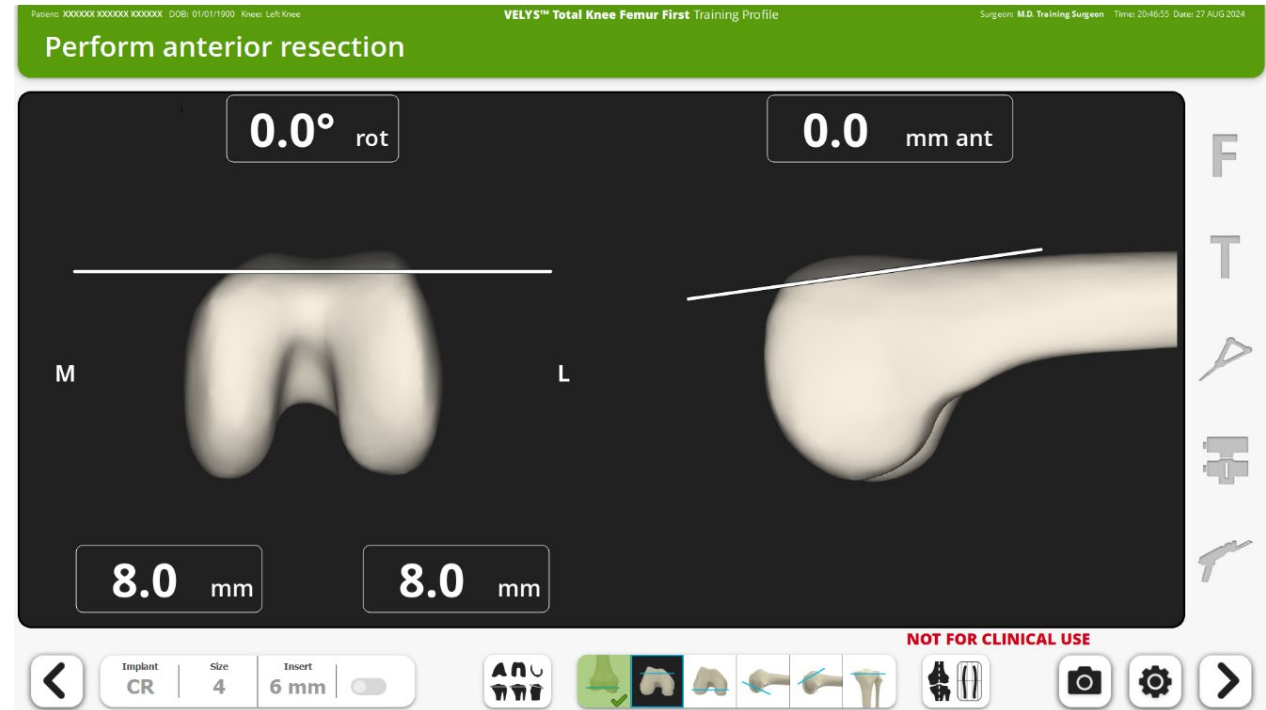
Plan Execution¹

Anterior Cut [varus knee]

Execute distal Anterior femur cut

Ensure accurate resection depth up to 0.5mm of accuracy by running the pointer along cut surface

Based on measurements and pointer verification, make any additional recuts to the anterior femoral cut



Plan Execution¹

Posterior Cut [varus knee]

Execute posterior femoral cut

Ensure accurate resection depth up to 0.5mm of accuracy by running the pointer along cut surface

Based on measurements and pointer verification, make any additional recuts to the posterior condyle cuts



Plan Execution¹

Proximal Tibia Cut [varus knee]

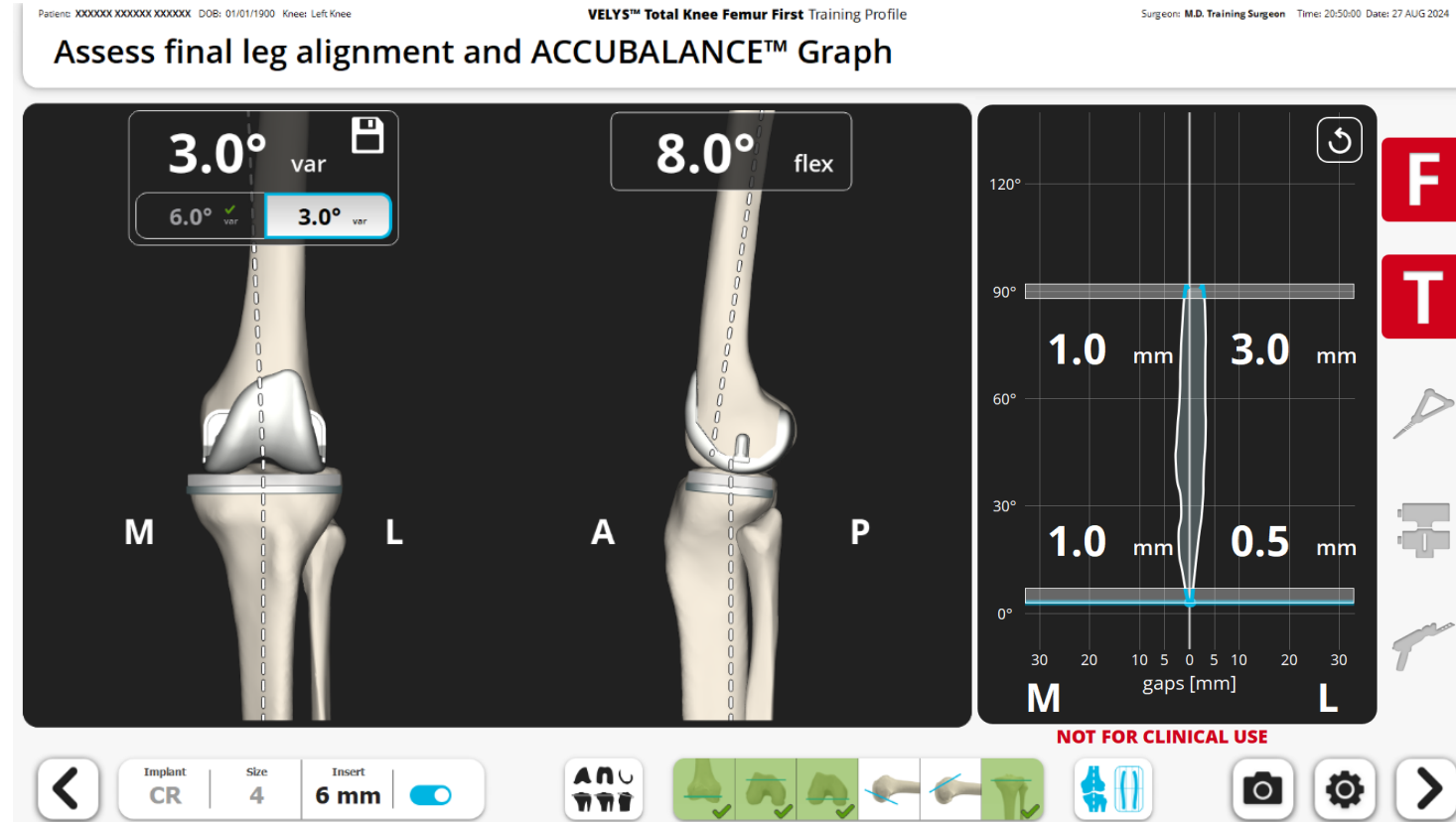
Execute proximal tibia cut

Ensure accurate resection depth up to 0.5mm of accuracy by running the pointer along cut surface

Based on measurements and pointer verification, make any additional recuts to the proximal tibia cut



Final leg alignment and ACCUBALANCE™ Graph¹



Inverse Kinematic Alignment with VELYS™ Robotic- Assisted Solution

Anatomic Variability⁴



■ KNEE

Coronal Plane Alignment of the Knee (CPAK) classification

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Methods

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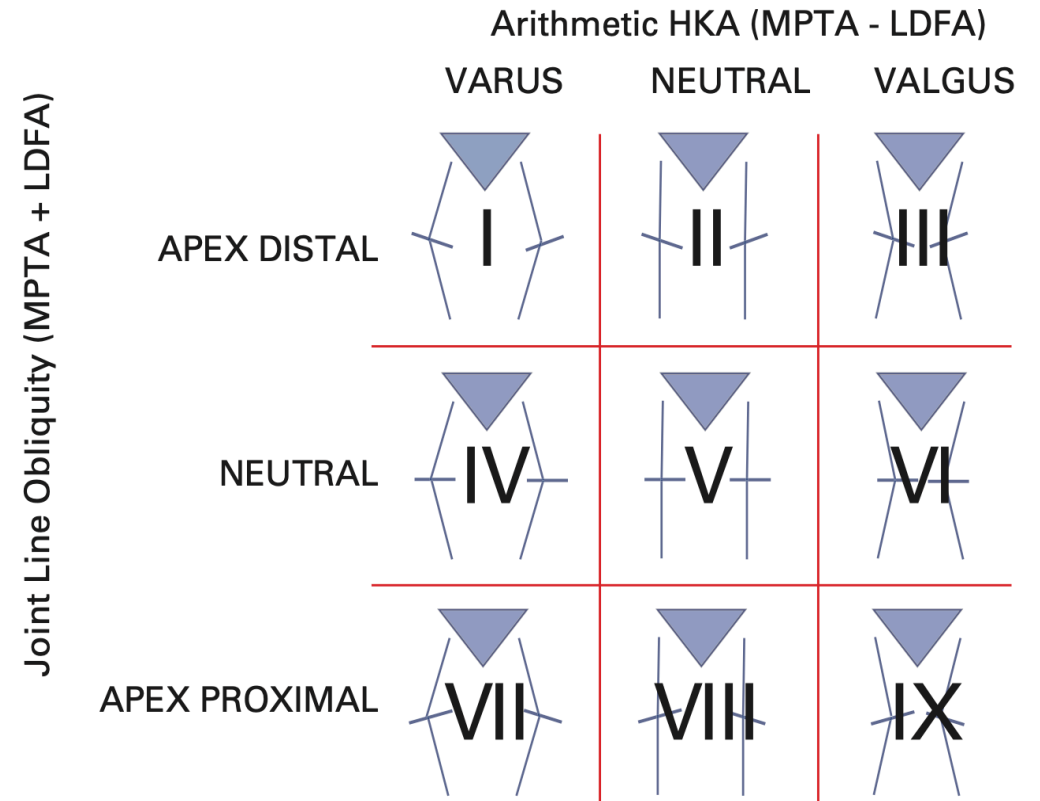
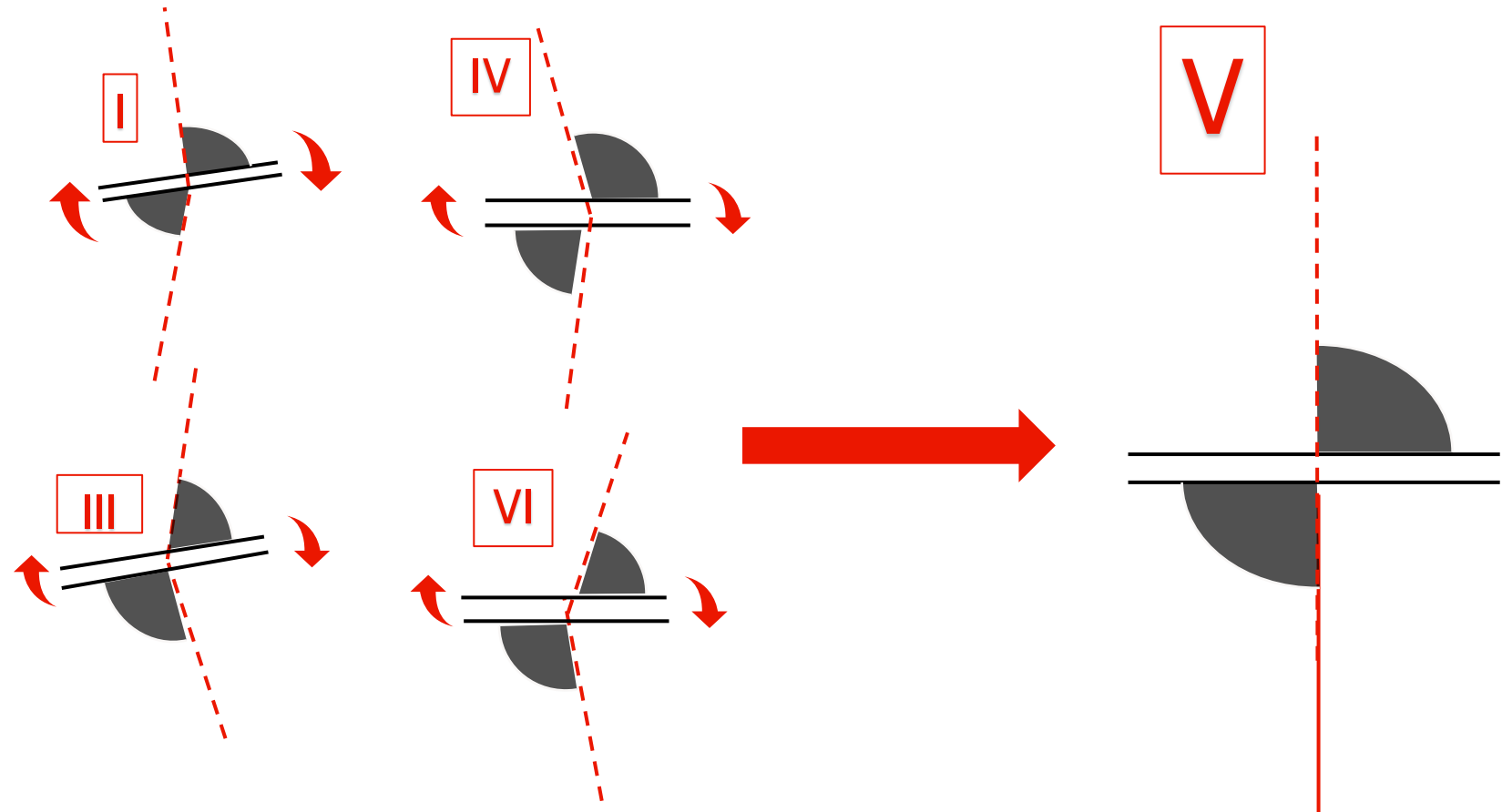
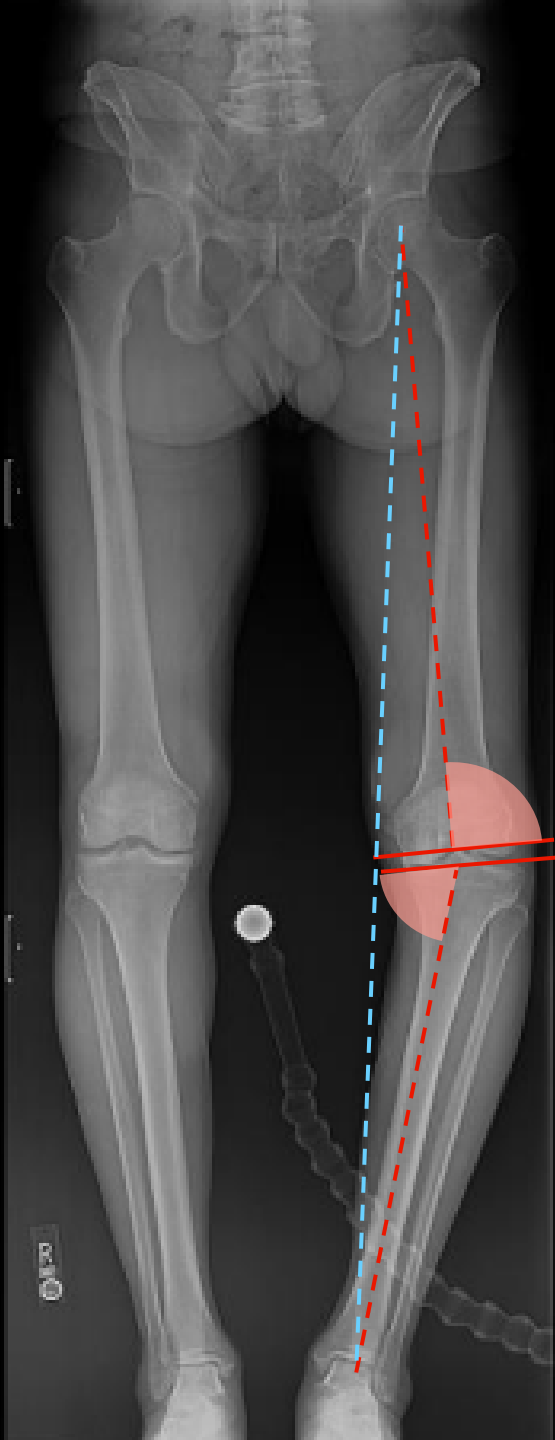


Fig. 4

Coronal Plane Alignment of the Knee classification (CPAK) with nine theoretical types of knee. Arithmetic HKA, arithmetic hip-knee-ankle angle.

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Historical techniques change native knee phenotypes



Native CPAK

Final CPAK

The major players...

Bony anatomy + Soft Tissues + Prosthesis

Inverse Kinematic Alignment Defined

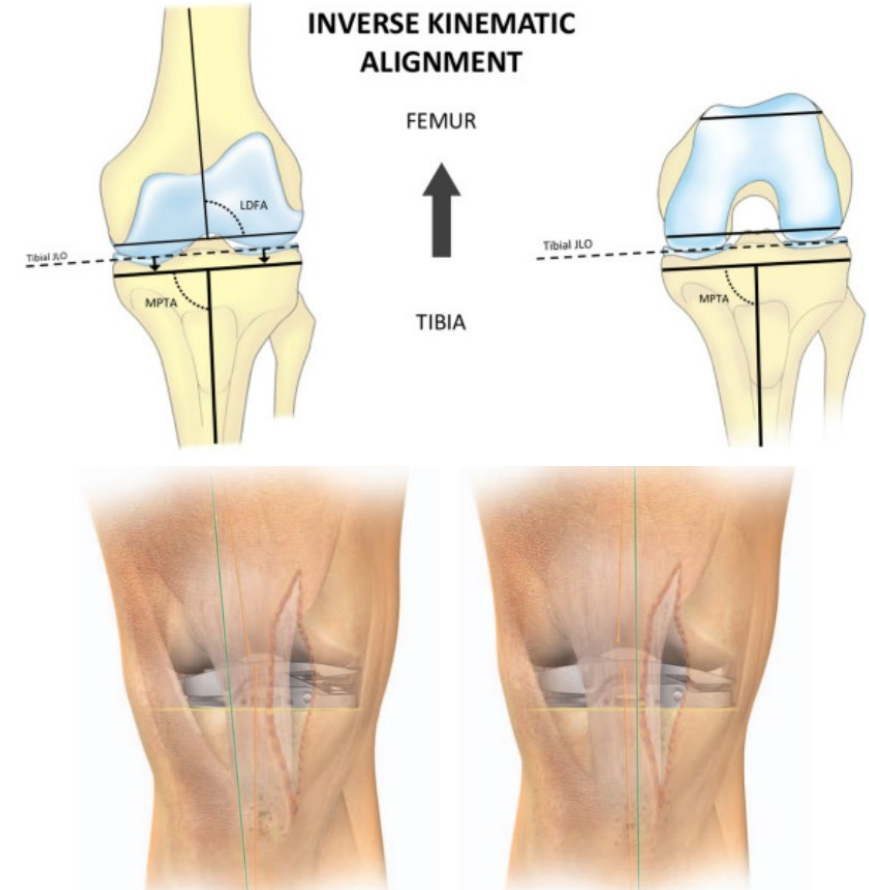
Goal: Restore pre-arthritic tibial joint line and adjust the ligament balance in the femoral distal and posterior bone cut to balance extension and flexion gaps independently

- **Restore normal kinematics in TKA**

- Interaction between ligaments, menisci, and articular surfaces of femur, tibia, and patella determine normal knee kinematics...multiplanar approach

How: Surgeon strives resurface tibial joint line making anatomic tibial cut, and then gap balances the knee within the boundaries of HKA angle between 174 and 183

- Resurface tibial joint line accounting for wear of cartilage and bone
- Apply tension throughout the range of motion
- Adjust femoral coronal and sagittal alignment to independently balance extension and flexion gaps



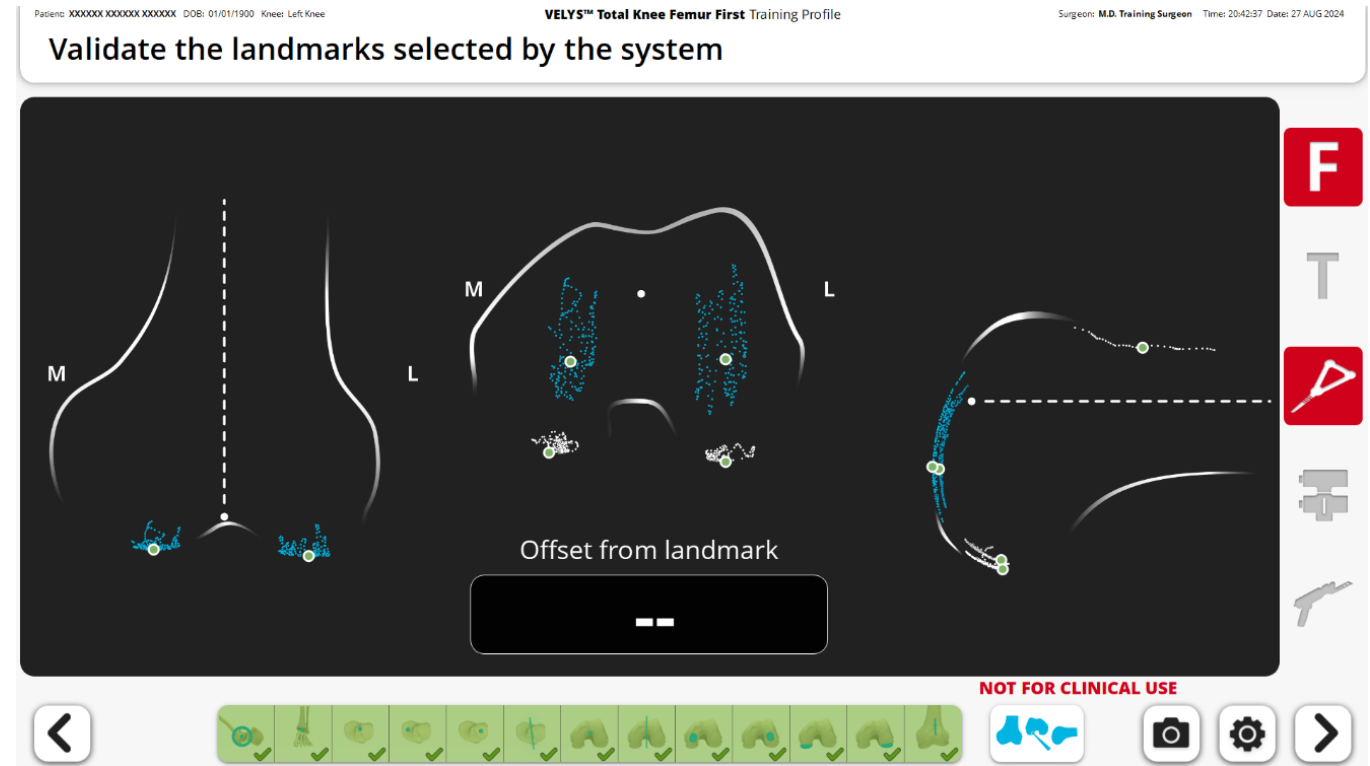
Landmark Acquisition – surgeon defined²

Surgeons dictate coronal plane resection depths and alignment through user defined registration

- Image based systems don't give the surgeon freedom to define origin of resection depths
 - CT scans must be segmented by third party

Surgeons will have ability to accommodate soft and hard tissue at their choosing

Surgeons can validate femoral registration within 0.5mm of accuracy



Landmark Acquisition Technique²

Define Femoral coordinate system

Femoral Head Center

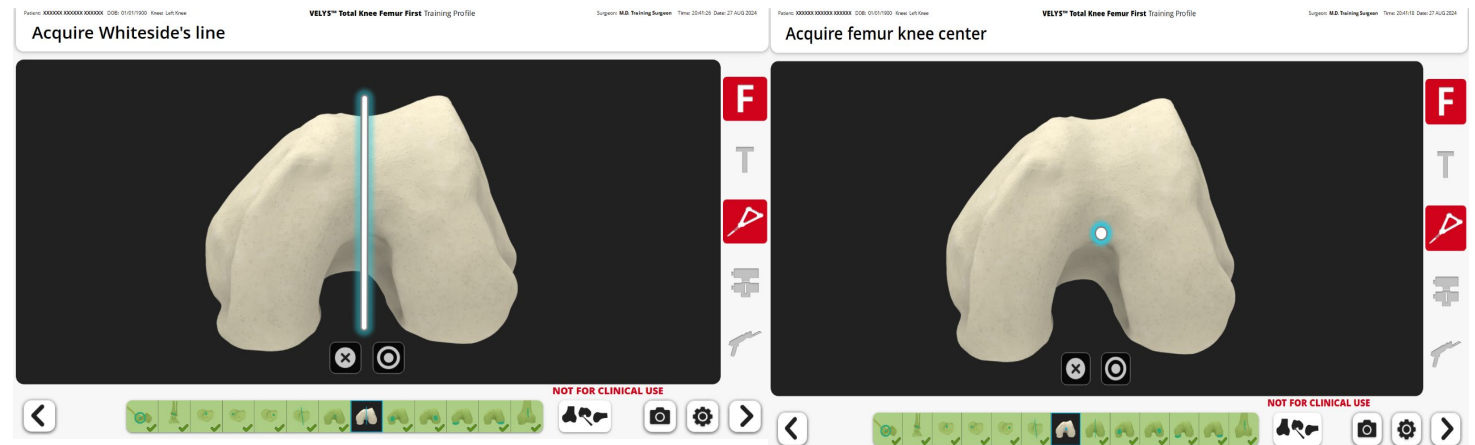
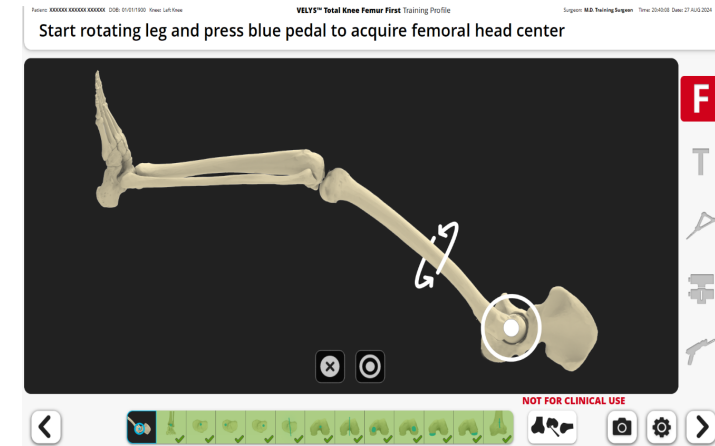
- Define proximal extent of femoral mechanical axis

Femoral Knee Center

- Define distal extent of femoral mechanical axis
- Apex of intercondylar notch

Whitesides Line (PCA 3 ER)

- Define secondary Int/Ext rotation reference for sagittal alignment



Landmark Acquisition Technique²

Define Tibial coordinate system

Medial & Lateral Malleoli

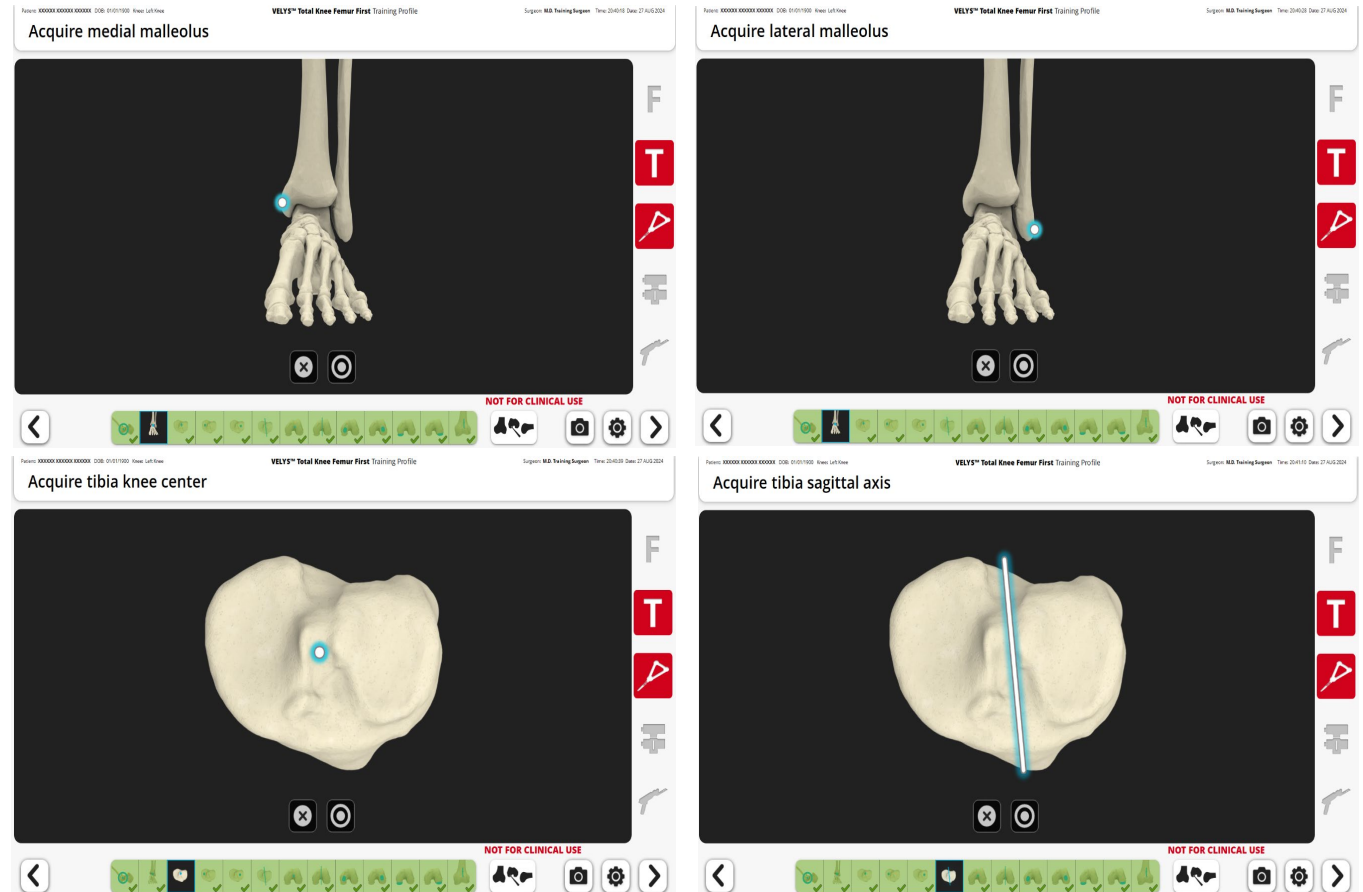
- Define distal extent of Tibial mechanical axis
- Palpate most prominent points on each malleoli

Tibial Knee Center

- Define proximal extent of Tibial mechanical axis
- Mid sagittal / Mid coronal in ACL footprint

Tibial Sagittal Axis

- Tip of the pointer aligned to PCL insertion posteriorly
- Shaft of pointer bisects tibial spine
- Defining A/P direction of slope

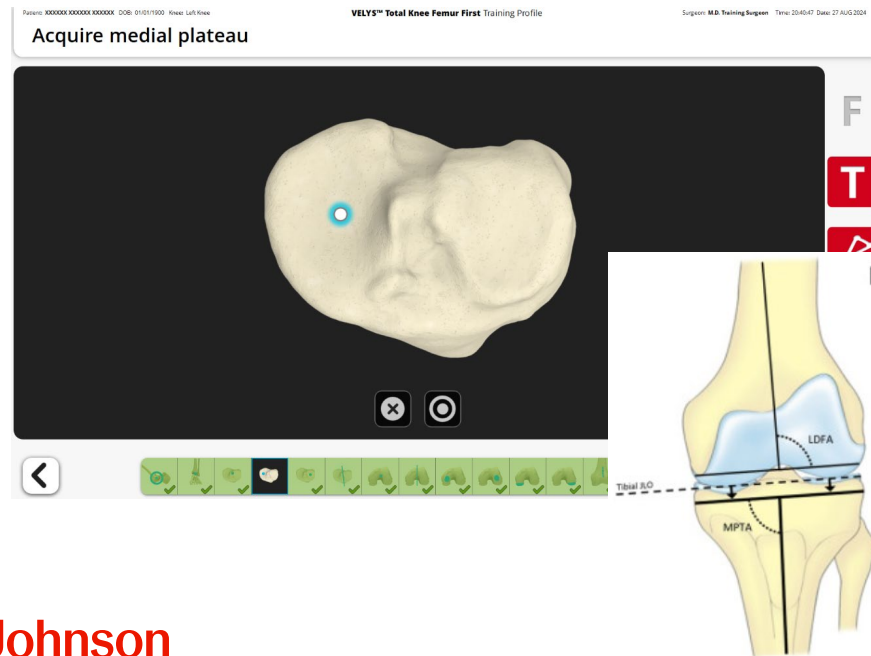


Landmark Acquisition Technique²

Define Tibial Resection Reference Points [varus knee]

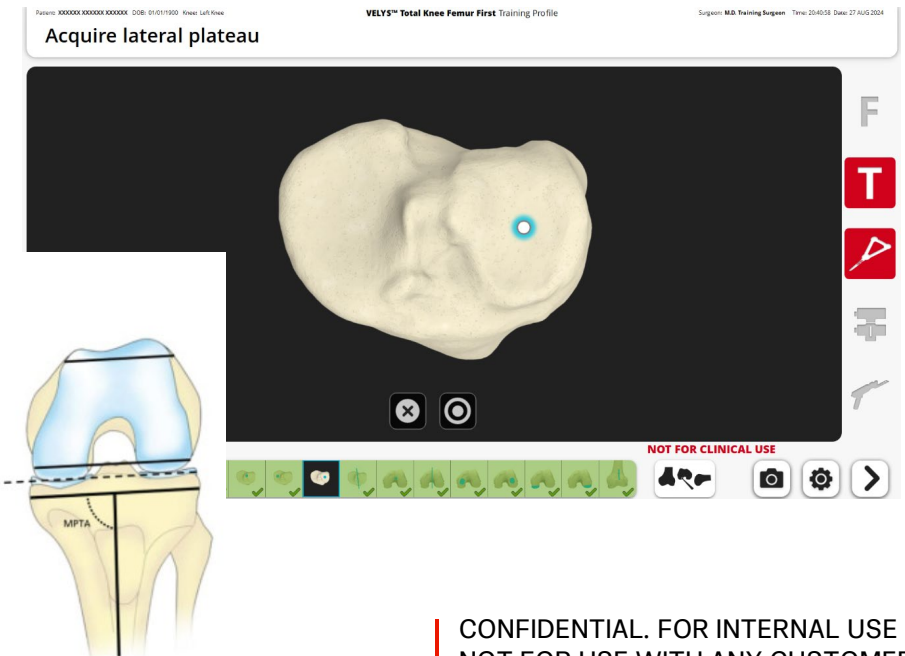
Medial Plateau

- Assess cartilage and bone loss/wear pattern
- Acquire medial resection reference at base of tibial spine
- Surgeon has flexibility to define where best to acquire this point based on their own opinion



Lateral Plateau

- Assess unworn plateau
- Acquire lateral resection reference at tide mark / base of tibial spine

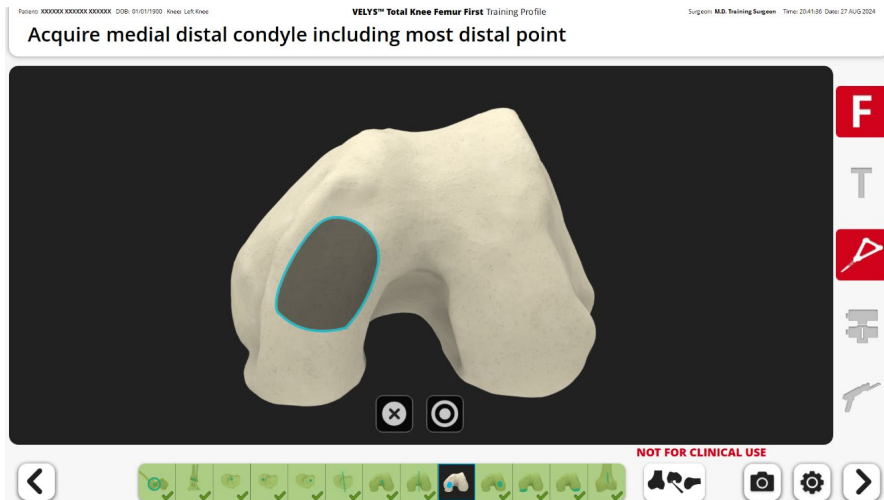


Landmark Acquisition Technique²

Define Coronal Plane [varus knee]

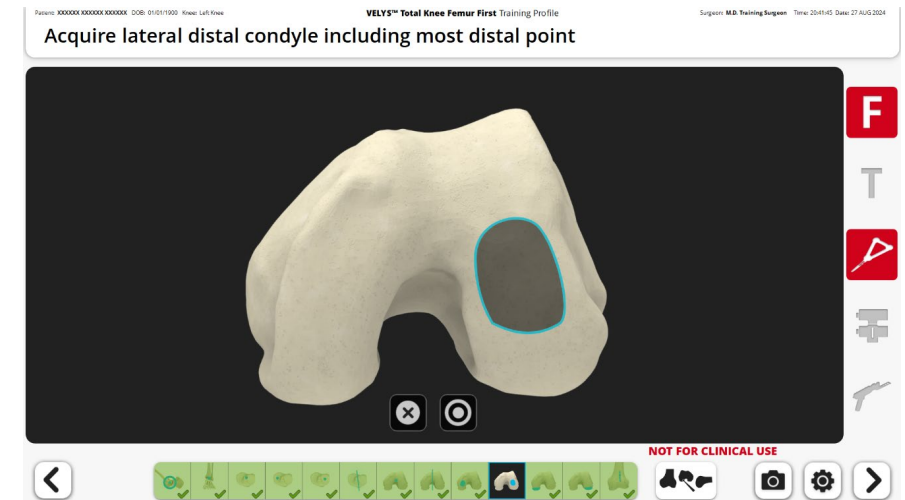
Medial Distal Femoral Condyle

- Define most distal point on medial femoral condyle
- Acquire most distal medial femoral condyle point through painting broad surface area



Lateral Distal Femoral Condyle

- Define most distal point on lateral femoral condyle
- Acquire most distal lateral femoral condyle point through painting broad surface area

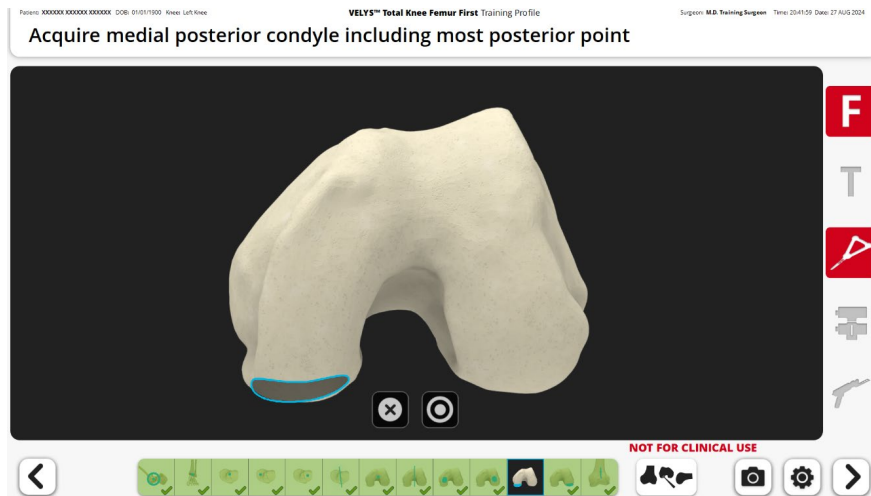


Landmark Acquisition Technique²

Define Femoral Sagittal Plane [varus knee]

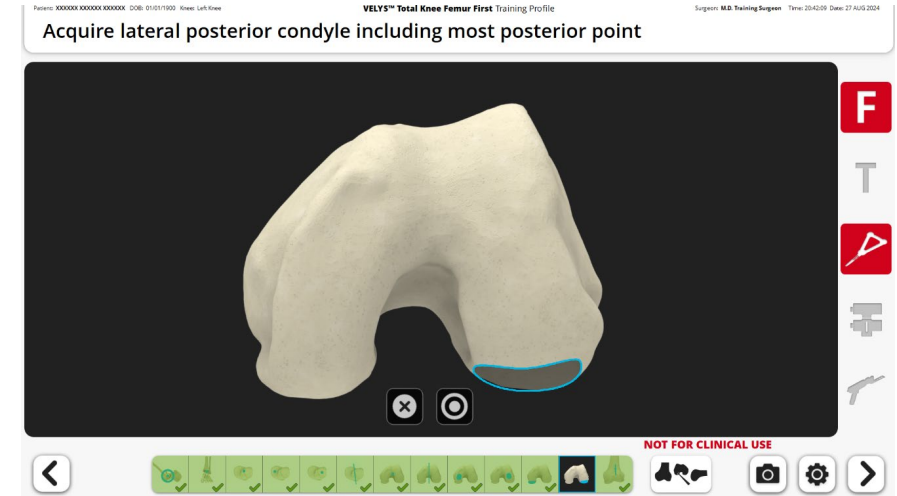
Medial Posterior Femoral Condyle

- Define most posterior medial femoral condyle point
- Acquire most posterior medial point through surface acquisition



Lateral Posterior Femoral Condyle

- Define most posterior lateral femoral condyle point
- Acquire most posterior lateral point through surface acquisition

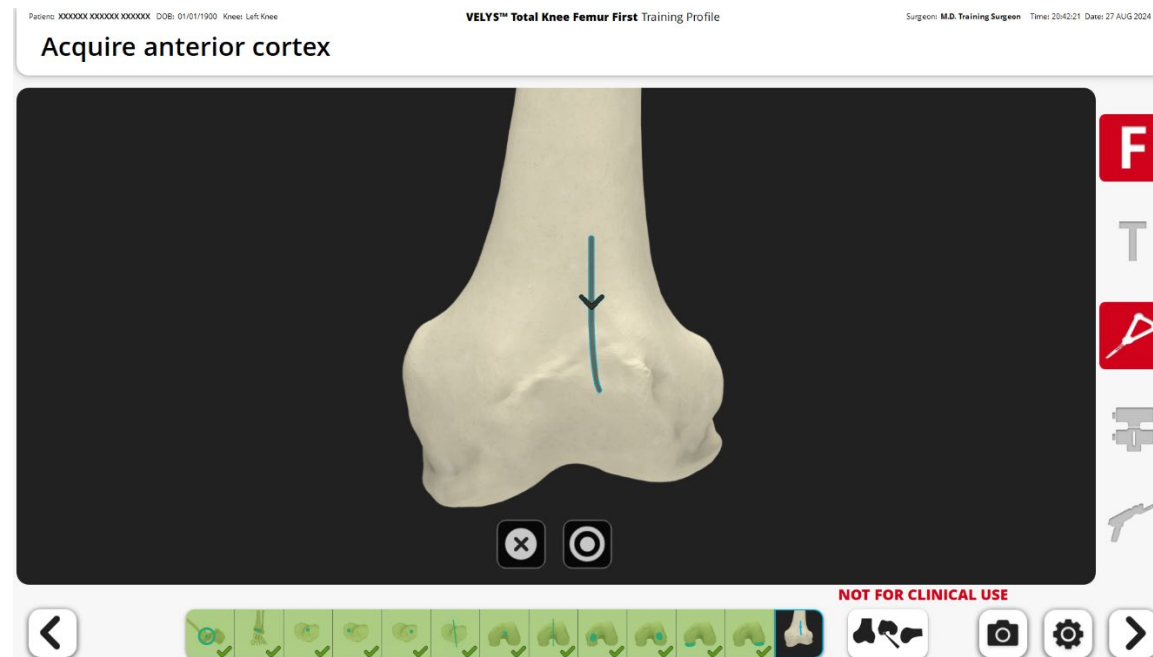


Landmark Acquisition Technique²

Define Femoral Sizing and Saw Blade Exit Point [varus knee]

Anterior Cortex Line

- Define femoral sizing and saw blade exit point
- Acquire anterior cortex line by running pointer in a straight line down the anterior cortex of the femur
 - Start pointer more proximal and run pointer down cortex about half way up the lateral ridge

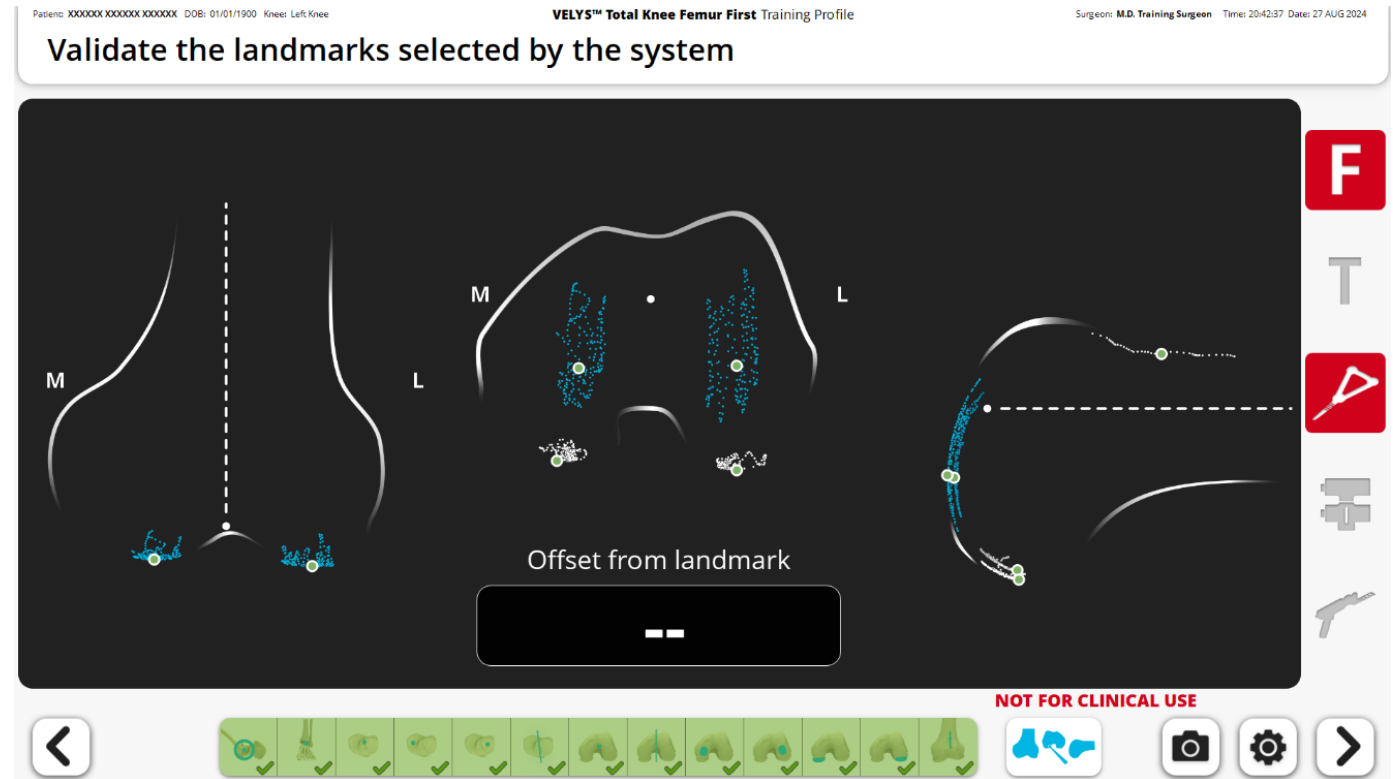


Landmark Acquisition Technique

Validate Femoral Landmarks [varus knee]

Surgeon defined landmark acquisition validation

- Confirm landmarks used to define femoral alignment in coronal plane (ext/flex)
- Surgeons can validate femoral registration up to 0.5 mm of accuracy
- If surgeon can't verify, surgeon has ability to re-acquire landmark
- Surgeon does NOT have to compromise their landmark acquisition based on pre-op imaging

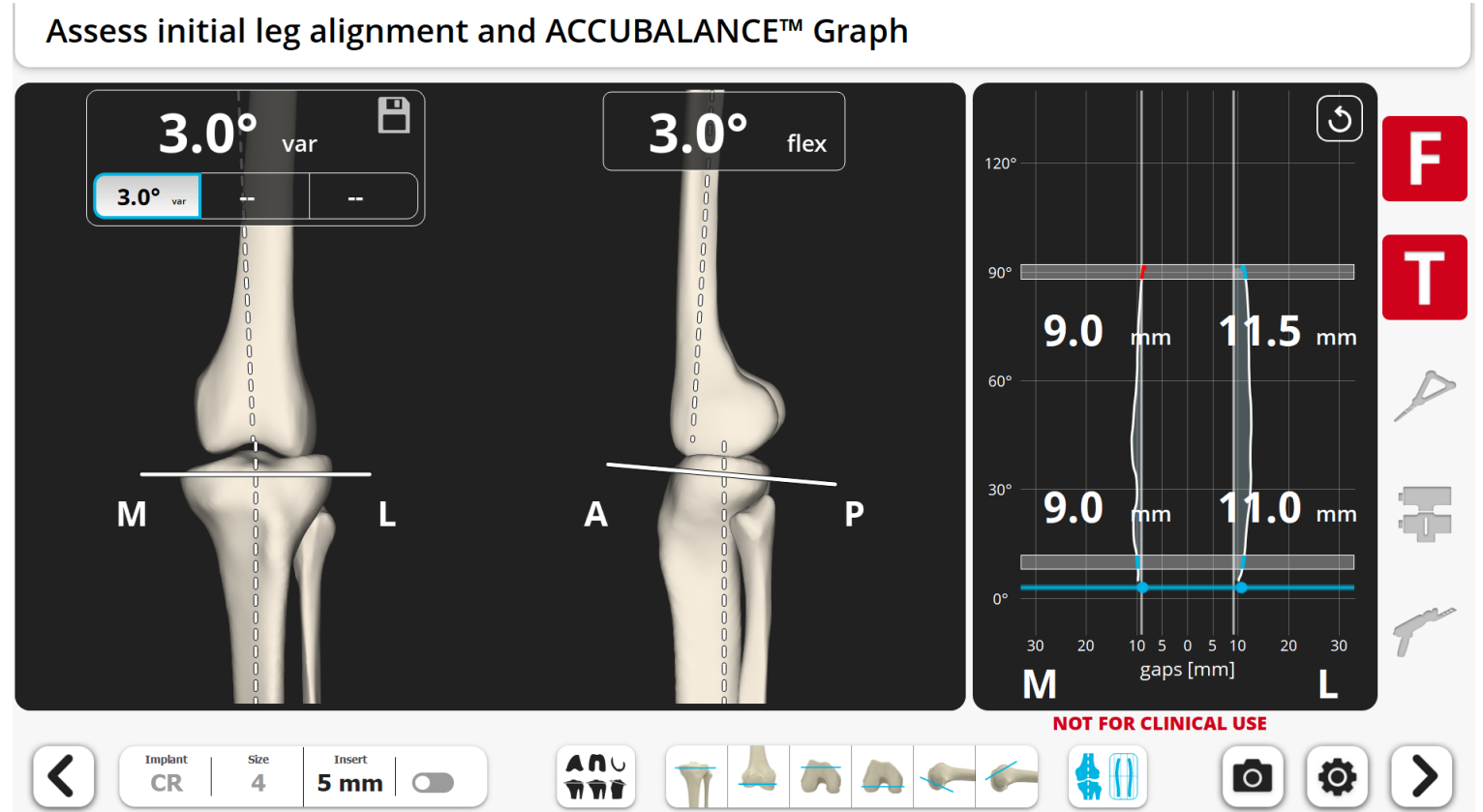


Initial Leg Alignment and ACCUBALANCE™ Graph²

Initial Leg Alignment and ACCUBalance Graph [varus knee]

Bring leg into full extension and capture initial HKA

First priority in iKA is to resurface tibial joint line obliquity so there is no need to stress the knee at this point



PROADJUST™ Planning Page

Inverse Kinematic Alignment Prioritization

iKA

MPTA

Soft-tissue
Balance

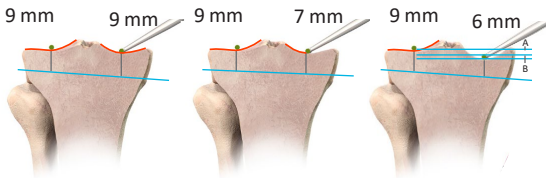
HKA
*if surgeon has
boundaries

LDFA

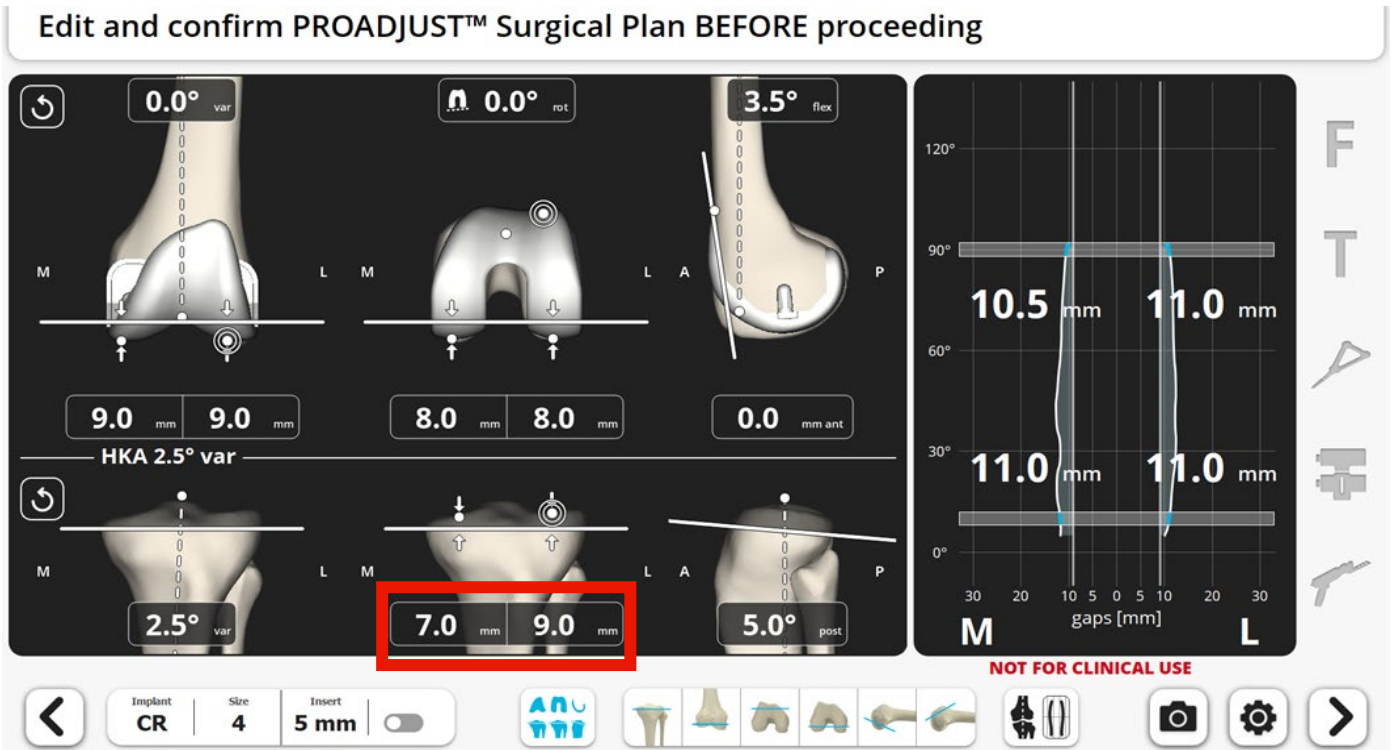
Medial Posterior
Femoral Condyle
Resection Depth

Lock resection laterally

Add varus to achieve
equal resections
medially and laterally



Scenario	1	2	3
Cartilage loss	0 mm	2 mm	2 mm
Bone loss	0 mm	0 mm	1 mm
Resection	9 mm	7 mm	6 mm
Total	9 mm	9 mm	9 mm



Landmark Acquisition Technique²

Proximal Tibia Cut [varus knee]

Execute proximal tibia cut

Ensure accurate resection depth up to 0.5mm of accuracy by running the pointer along cut surface

Based on measurements and pointer verification, make any additional recuts to the proximal tibia cut

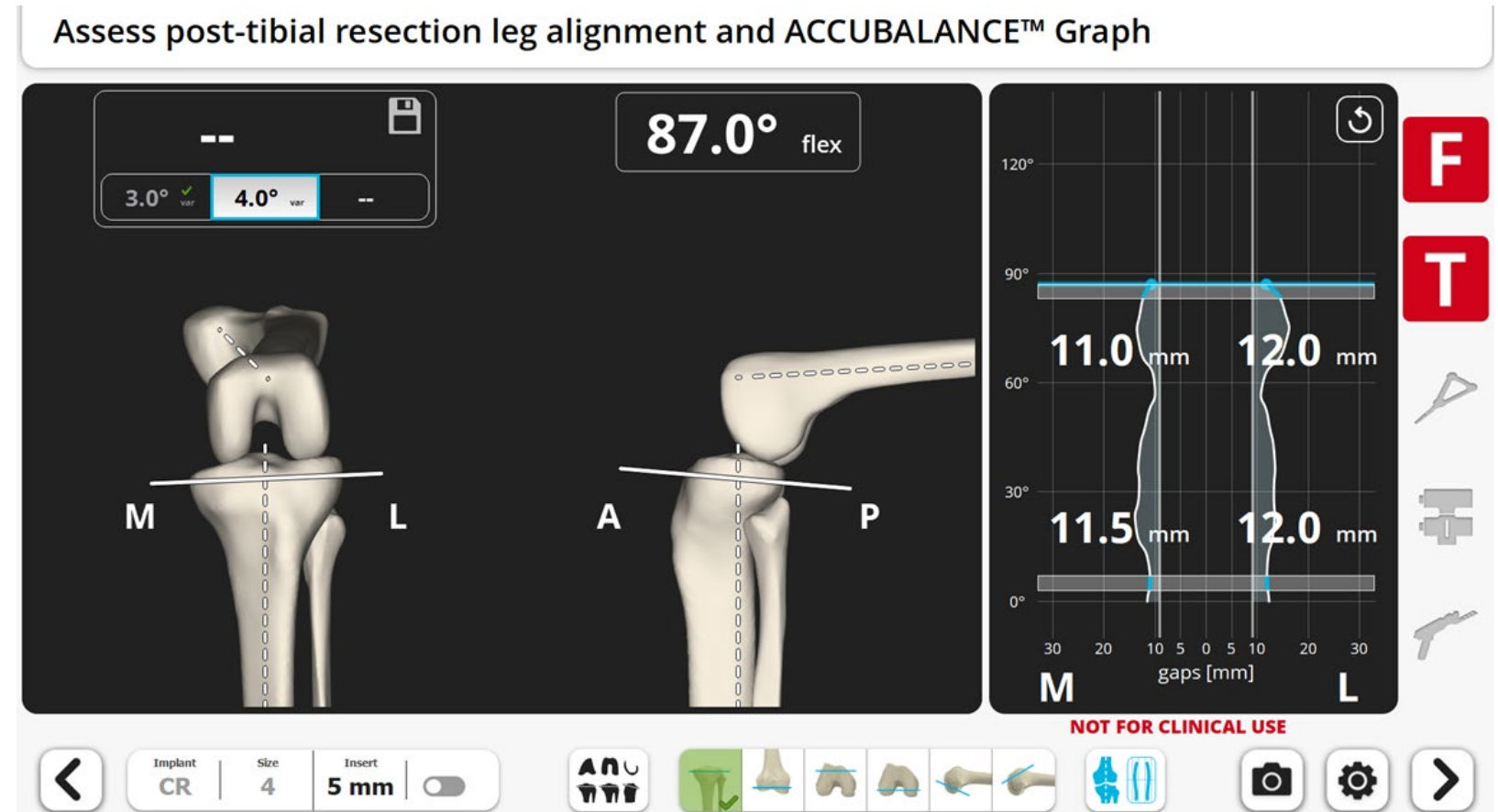


Tensioned Leg Alignment & ACCUBALANCE™ Graph²

Introduce soft tissue tensioner

Take leg through ROM allowing tensioner to consistently tension ligaments throughout ROM

Assess ACCUBALANCE™ Graph to determine if Extension Gap Window should be moved to avoid planning femoral adjustments off of inaccurate data



PROADJUST™ Planning Page

Post Tibial Resection – tensioner introduced [varus knee]

iKA

MPTA

Soft-tissue
Balance

HKA
*if surgeon has
boundaries

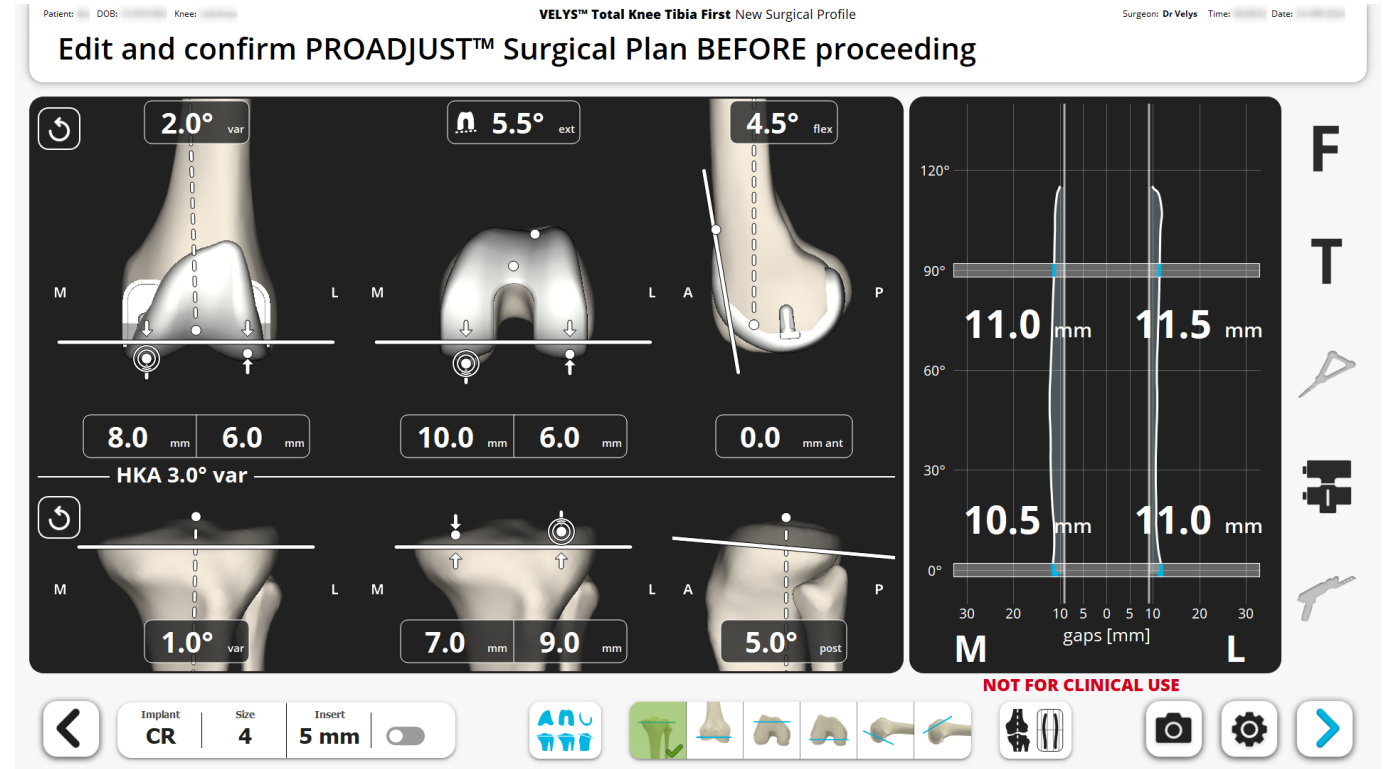
LDFA

Medial Posterior
Femoral Condyle
Resection Depth

Allow soft tissue data to inform
coronal and sagittal alignment
adjustments made to femur

Target equal gaps in extension and
medially in flexion allowing for laxity
laterally in flexion

Once gaps are balanced, asses
LDFA and medial posterior femoral
condyle resection depth to
determine if appropriate ~w/in 1 mm



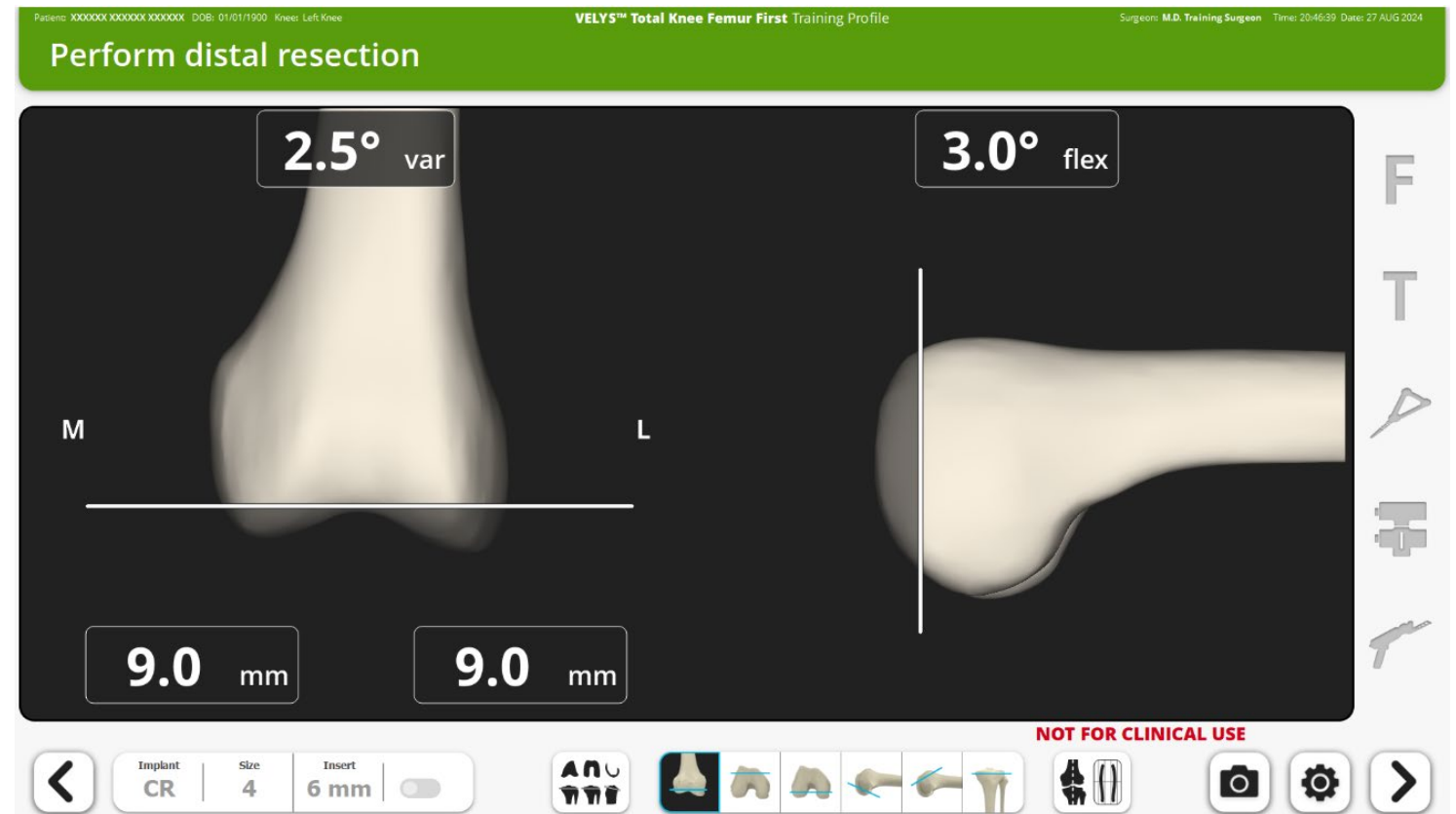
Plan Execution²

Distal Cut [varus knee]

Execute distal femoral cut

Ensure accurate resection depth up to 0.5mm of accuracy by running the pointer along cut surface

Based on measurements and pointer verification, make any additional recuts to the distal femoral condyle cuts



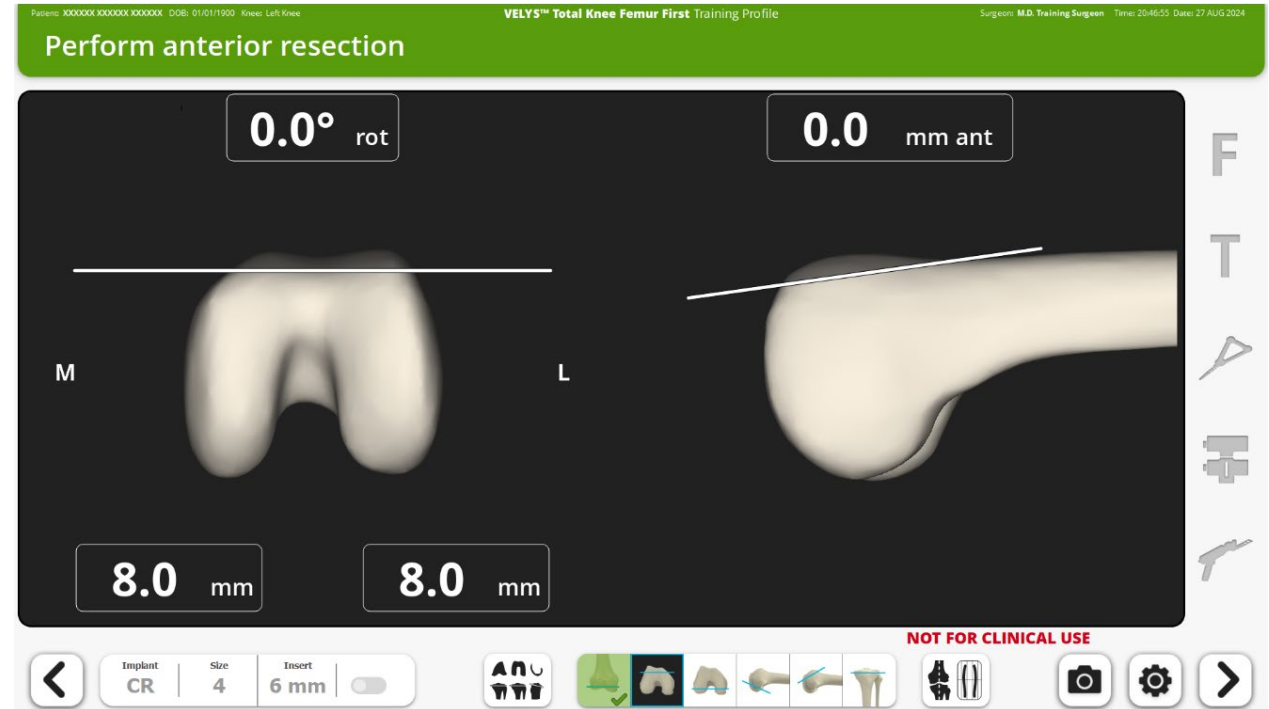
Plan Execution²

Anterior Cut [varus knee]

Execute distal Anterior femur cut

Ensure accurate resection depth up to 0.5mm of accuracy by running the pointer along cut surface

Based on measurements and pointer verification, make any additional recuts to the anterior femoral cut



Plan Execution²

Posterior Cut [varus knee]

Execute posterior femoral cut

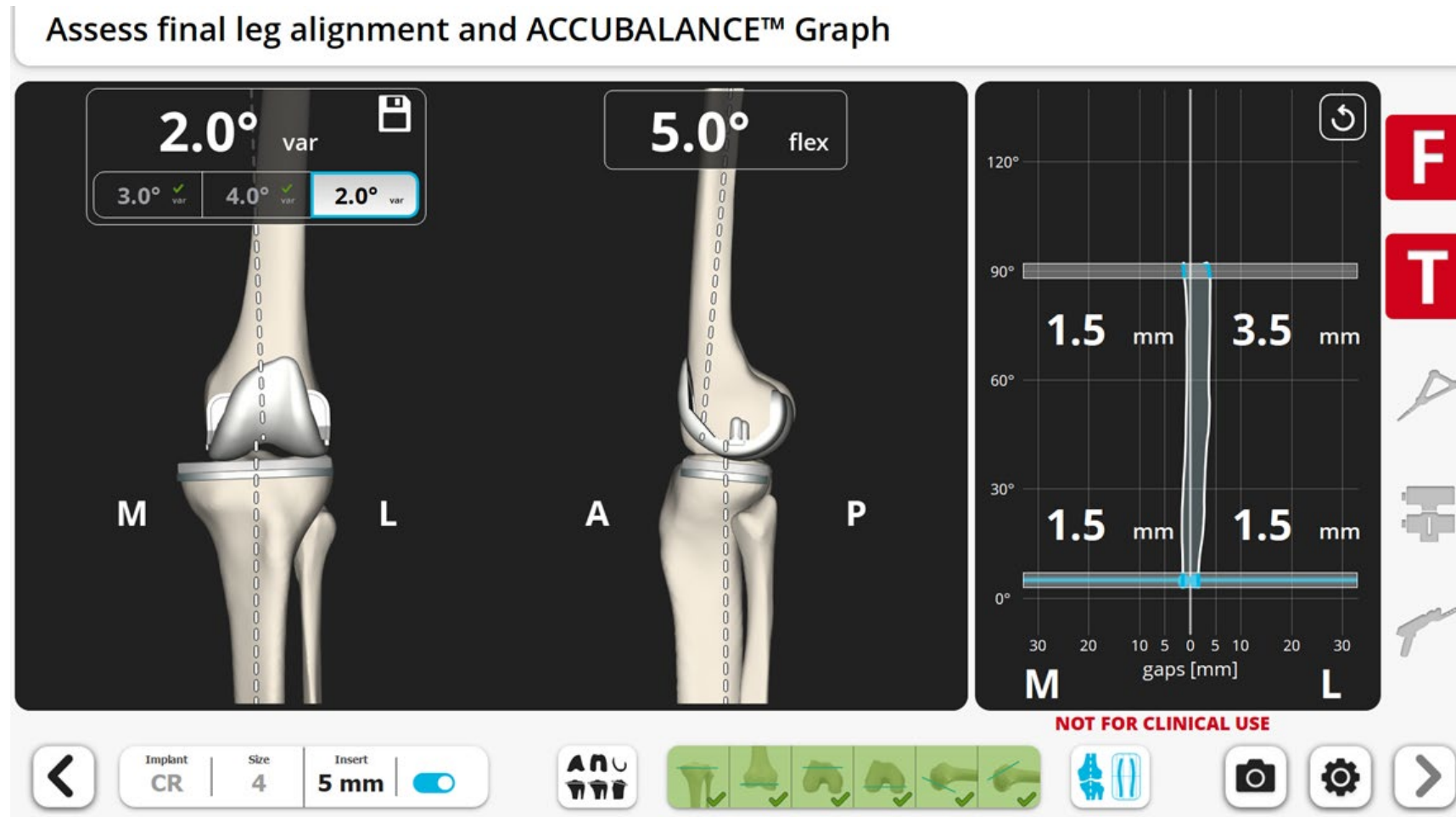
Ensure accurate resection depth up to 0.5mm of accuracy by running the pointer along cut surface

Based on measurements and pointer verification, make any additional recuts to the posterior condyle cuts



Final leg alignment and ACCUBALANCE™ Graph²

Final Leg Alignment and ACCBALANCE™ Graph [varus knee]

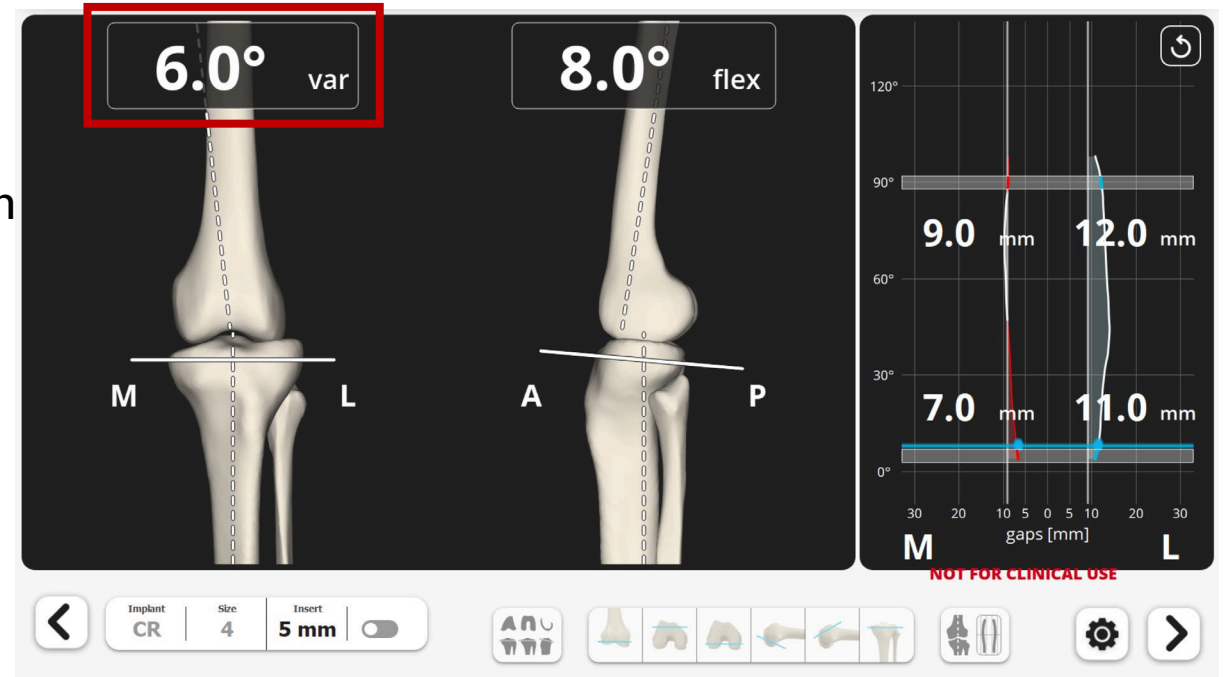


Initial Leg Alignment & ACCUBALANCE™ Graph²

Priority – Long Leg Alignment [Target HKA]

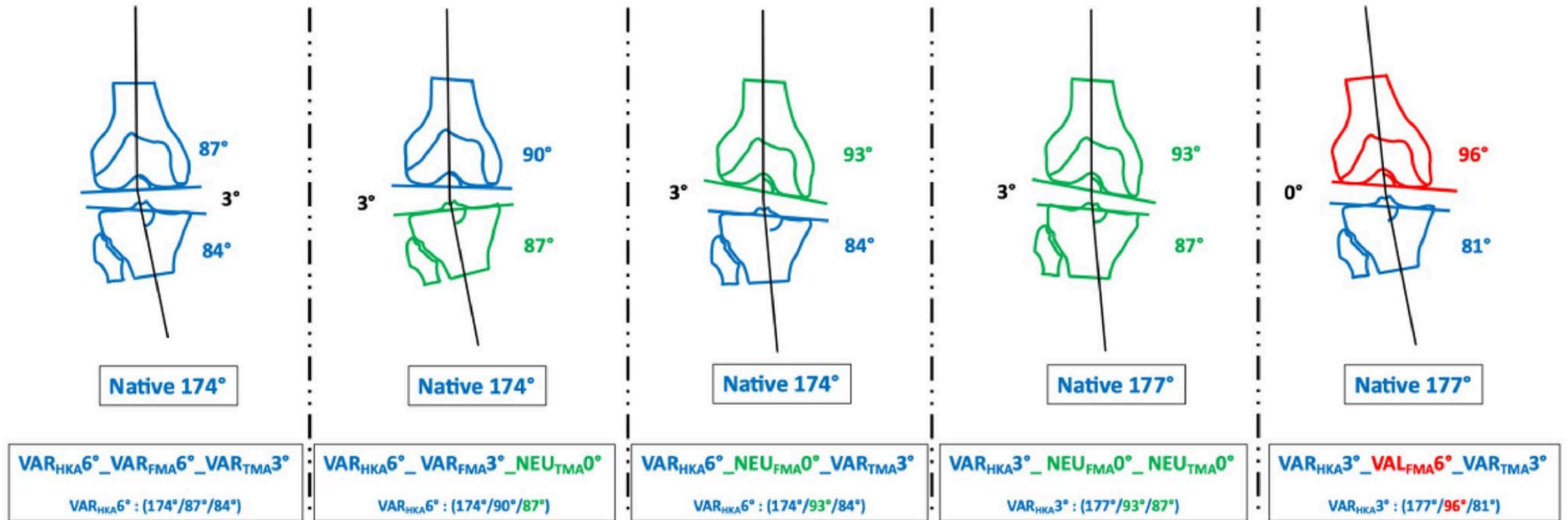
Femur First Technique of resurfacing the femur based off registration then cutting the tibia with respect to surgeon priority:

- Long Leg Alignment [Target HKA]
 - Apply valgus stress to corrected Varus alignment
 - Corrected Varus alignment becomes Target HKA on PROADJUST™ Planning Page



Varus Phenotypes⁶

per Dr. Hirschmann

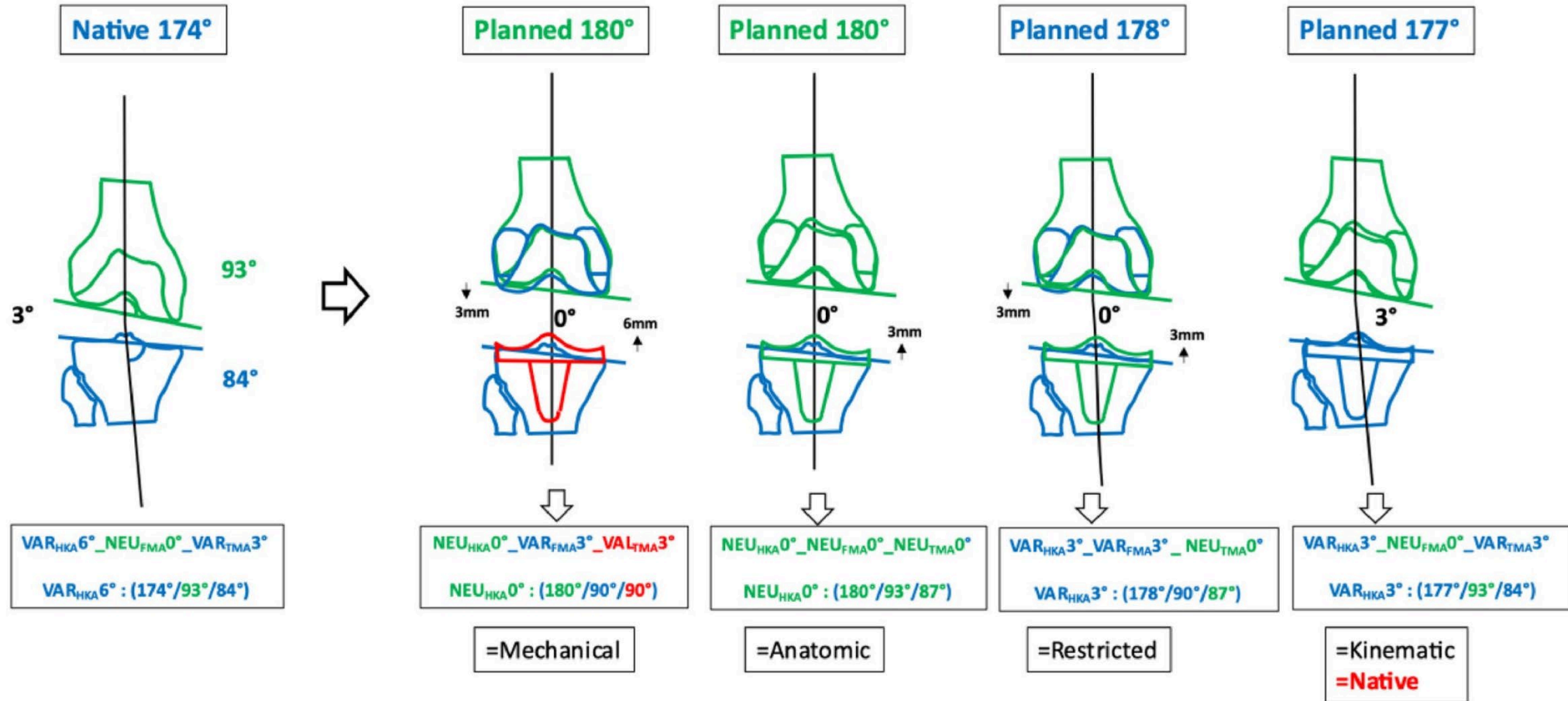


Reference: Schelker, Benjamin & Moret, Céline & Sava, Manuel-Paul & Eisenhart-Rothe, Rüdiger & Graichen, Heiko & ARNOLD, Markus & Vincent, Leclercq & Hirschmann, Michael. (2023). The impact of different alignment strategies on bone cuts in total knee arthroplasty for varus knee phenotypes. Knee surgery, sports traumatology, arthroscopy : official journal of the ESSKA. 31. 10.1007/s00167-023-07351-w.

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VAR_{HKA} 6°_NEU_{FMA} 0°_VAR_{TMA} 3° [174°/93°/84°]⁶

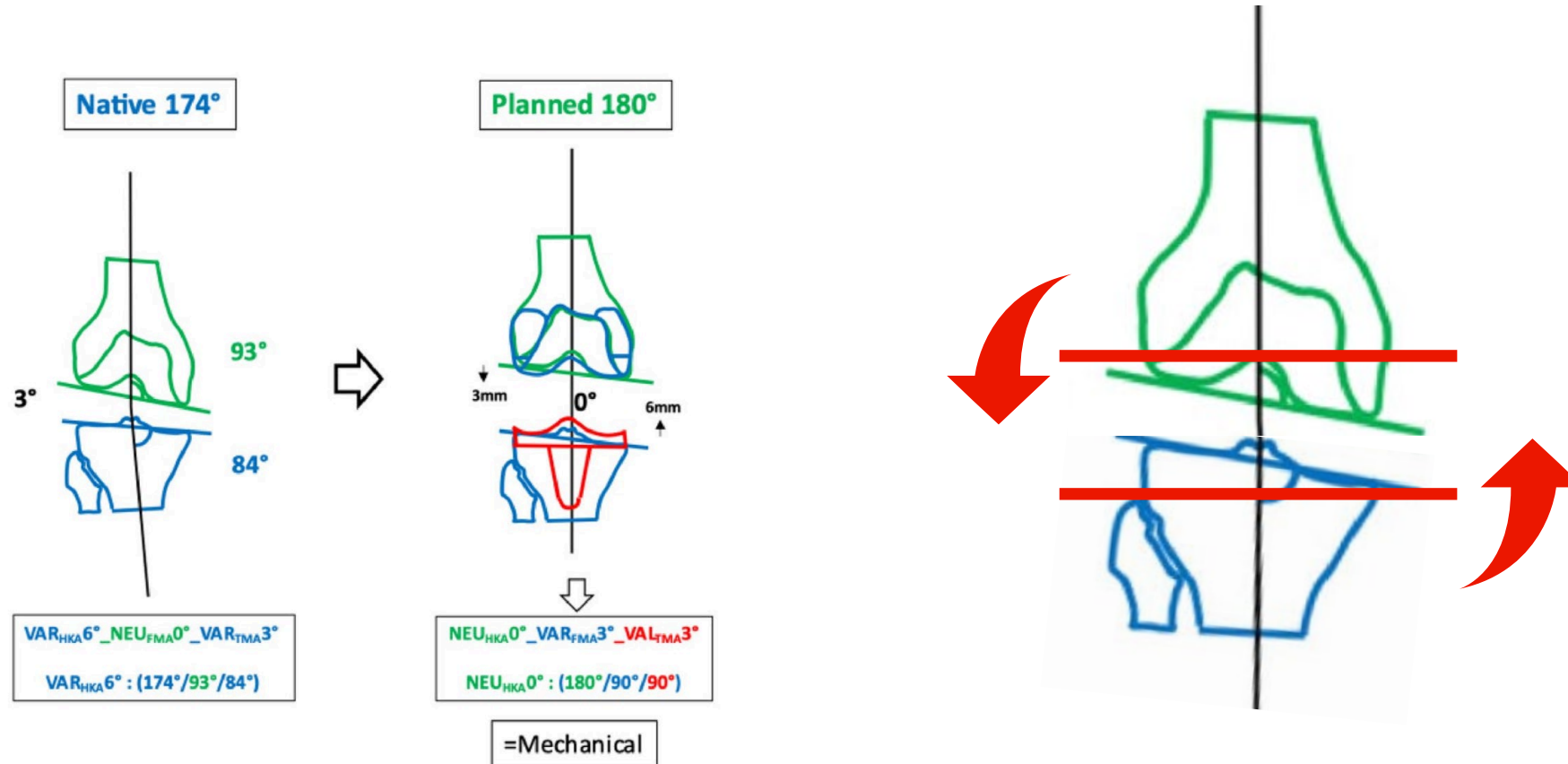
changes per alignment philosophy



Reference: Schelker, Benjamin & Moret, Céline & Sava, Manuel-Paul & Eisenhart-Rothe, Rüdiger & Graichen, Heiko & ARNOLD, Markus & Vincent, Leclercq & Hirschmann, Michael. (2023). The impact of different alignment strategies on bone cuts in total knee arthroplasty for varus knee phenotypes. Knee surgery, sports traumatology, arthroscopy : official journal of the ESSKA. 31. 10.1007/s00167-023-07351-w.

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$VAR_{HKA} 6^{\circ} - NEU_{FMA} 0^{\circ} - VAR_{TMA} 3^{\circ} [174^{\circ}/93^{\circ}/84^{\circ}]^6$ changes per alignment philosophy



Reference: Schelker, Benjamin & Moret, Céline & Sava, Manuel-Paul & Eisenhart-Rothe, Rüdiger & Graichen, Heiko & ARNOLD, Markus & Vincent, Leclercq & Hirschmann, Michael. (2023). The impact of different alignment strategies on bone cuts in total knee arthroplasty for varus knee phenotypes. Knee surgery, sports traumatology, arthroscopy : official journal of the ESSKA. 31. 10.1007/s00167-023-07351-w.

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